

Omni-Embed-Mini: Binding Modalities Without Forgetting via Dense Distillation

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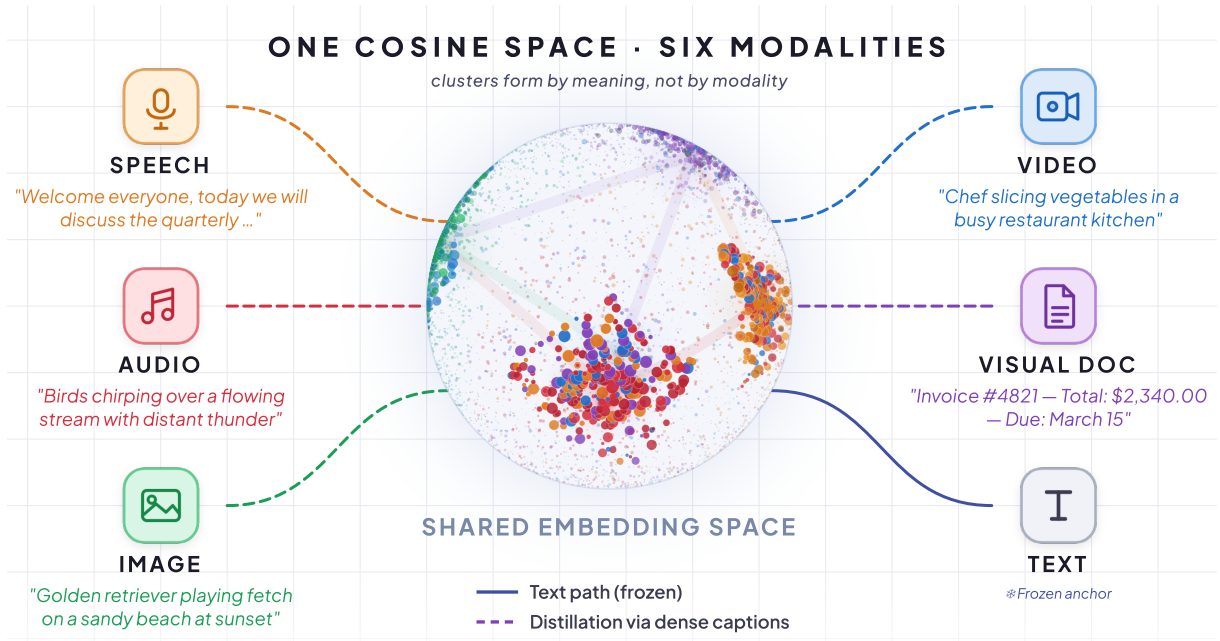


Figure 1: **Omni-Embed-Mini overview.** Omni-Embed-Mini maps speech, audio, image, video, visual-document, and text inputs into a single cosine embedding space, where semantically related examples cluster together regardless of modality. The model keeps the text encoder as a frozen anchor and aligns non-text modalities to this shared space through distillation from dense captions.

Abstract

Extending a text embedding model to new modalities typically *degrades* text retrieval quality, and existing omni-modal embedders compensate with multi-billion parameters. We present **Omni-Embed-Mini**, a 0.9B-parameter model that maps text, speech, audio, images, video, and visually-rich documents into a single shared cosine space *without* updating any text-side parameter. Our key insight is that the teacher signal requires no separate embedding model: each media sample is paired with a dense cascaded caption, and the teacher target is simply the frozen backbone’s own embedding of that caption. Because teacher and student share the same backbone weights, they inhabit byte-identical geometry, and lightweight projectors plus phased LoRA adapters on the

modality encoders suffice for alignment. Training combines a Matryoshka SigLIP contrastive loss with an online hybrid hard-negative miner whose negatives sharpen as the encoder improves. The recipe carries over to a 2.3B variant by swapping in a native vision-language backbone. Omni-Embed-Mini-0.9B keeps its text weights bit-identical to the backbone, so training *cannot* regress text retrieval (49.57 nDCG@10 on MTEB-v2 BEIR-8), while extending it to five additional modalities, and is $\sim 2.7\times$ to $9.5\times$ smaller than every open omni embedder we compare against. The 2.3B variant is competitive with the closed gemini-embedding-2, edging ahead of it on the overall-modality average. Models, code, data and evaluation harness are on our [project page](#).

1 Introduction

A central goal of recent representation learning is a unified embedding space that binds diverse modalities (Girdhar et al., 2023; Zhu et al., 2024). In practice, most attempts to jointly contrastive-train text with vision and audio face two coupled failure modes. (i) *Catastrophic forgetting on the text side* (McCloskey and Cohen, 1989): fine-tuning a strong text embedder against media inputs shifts the text geometry and degrades MTEB performance. Text is still the most critical modality for real-world search, making this degradation highly undesirable. (ii) *Modality bloat*: existing omni embedders such as LCO-Embedding-Omni-3B/7B (Xiao et al., 2026), BidirLM-Omni-2.5B (Boizard et al., 2026), NVIDIA’s omni-embed-nemotron-3B (Xu et al., 2025), and e5-omni-3B/7B (Chen et al., 2026) have scaled to 2.5B to 8.9B parameters in search of enough capacity to absorb every modality, making them less suited to edge-deployed dynamic retrieval at scale. We term these two coupled failure modes the *multimodal expansion dilemma*: under current recipes, every additional modality forces a trade-off, because mitigating text-side catastrophic forgetting has typically come with much larger models.

We argue that text should not be a co-learner; it should be an *immovable anchor*. BLIP-2 (Li et al., 2023) bridges a frozen image encoder and a frozen LLM with a small Q-Former, and the first stage of LLaVA (Liu et al., 2023) aligns vision to a frozen LLM through a lightweight projector alone. We translate this principle to retrieval, drawing on the same logic Reimers and Gurevych (2020) used to align a new language to a frozen English teacher’s sentence space: each media sample’s teacher target is the EOS-pooled embedding of a dense cascaded caption, computed by the *identical frozen backbone* that processes the student. Both targets and text queries are therefore produced by byte-identical weights; a modest projector and phased LoRA on the encoders close the modality gap. We call the resulting recipe **Omni-Embed-Mini**.

As a result, Omni-Embed-Mini offers a favourable size-vs-performance trade-off. Despite being $\sim 2.7\times$ to $9.5\times$ smaller than the open omni-modal embedders listed above, our 0.9B model leaves its foundational text retrieval path untrained and covers five additional modalities in a single sub-billion-parameter model; on media retrieval it trails these larger baselines, as ex-

pected at its size. Our 2.3B variant is ahead of the closed gemini-embedding-2 on the overall-modality average (51.39 against 49.51), leading on image, video and visual documents while trailing on text, speech and audio.

Our contribution is not a new alignment primitive. We combine dense captioning, frozen-backbone self-distillation, parameter-efficient modality adaptation, Matryoshka training and on-line mining into an omni-modal retrieval recipe with one specific invariant: the deployed text inference path is unchanged. Concretely: (i) **Zero-regression omni embedding**: a 0.9B model (Table 1) scores 49.57 nDCG@10 on MTEB-v2 BEIR-8 while adding five modalities; the text branch’s weights are bit-identical to the stock 0.6B backbone, so training cannot regress it. (ii) **Cascaded-caption self-distillation through a shared backbone**: teacher and student share weights, so cosine alignment in the native pooled hidden space suffices: no projection head, no separate embedding teacher, fully cacheable targets. (iii) **Matryoshka SigLIP + online hybrid hard-negative mining + phased LoRA**: a parameter-efficient recipe that carries over to a 2.3B variant by swapping the backbone for a native vision-language model. (iv) To our knowledge, this is the **smallest open omni embedder** at the time of writing, covering text, speech, audio, image, video, and visually-rich documents in one cosine space.

2 Related Work

Per-modality embedders. *Text.* The MTEB benchmark family (Muennighoff et al., 2023; Thakur et al., 2021) has strong text encoders such as Qwen3-Embedding (Zhang et al., 2025b), voyage-3-m-exp (Voyage AI, 2025), Conan-embedding-v2 (TencentBAC, 2025), GritLM (Muennighoff et al., 2025), inf-retriever (Yang et al., 2025), and LENS (Lei et al., 2025); we treat these as text-side reference points, while our zero-regression claim is measured against our own frozen backbones: Qwen3-Embedding-0.6B for the 0.9B, and Qwen3-VL-Embedding-2B, including its vision paths, for the 2.3B. *Vision-language.* VLM2Vec-V2 (Meng et al., 2025), Qwen3-VL-Embedding (Li et al., 2026), RzenEmbed (Jian et al., 2025), Ops-MM-embedding (OpenSearch-AI Team, Alibaba Cloud, 2025), and seed-1.6 (ByteDance Seed, 2025) adapt pretrained vision-language models for retrieval, usually with contrastive training, but do not handle

Model	Backbone	Inference	Frozen	Trained	Trained %
Omni-Embed-Mini-0.9B	Qwen3-Embedding-0.6B	935.35 M	869.98 M	68.32 M	7.3%
Omni-Embed-Mini-2.3B	Qwen3-VL-Embedding-2B	2.44 B	2.30 B	141.68 M	5.8%
BidirLM-Omni-2.5B	Qwen3-1.7B $\times$ 3 merged	2.45 B	724 M	1.72 B	70.4%
omni-embed-nemotron-3B	Qwen2.5-Omni-3B Thinker	4.70 B	4.67 to 4.70 B	7 to 30 M	0.2 to 0.6%
e5-omni-3B	Qwen2.5-Omni-3B Thinker	4.70 B	4.58 to 4.70 B	7 to 120 M	0.2 to 2.5%
LCO-Embedding-Omni-3B	Qwen2.5-Omni-3B Thinker	4.70 B	4.58 B	120 M	2.5%
e5-omni-7B	Qwen2.5-Omni-7B Thinker	8.93 B	8.77 to 8.92 B	10 to 161 M	0.1 to 1.8%
LCO-Embedding-Omni-7B	Qwen2.5-Omni-7B Thinker	8.93 B	8.77 B	161 M	1.8%

Table 1: **Inference, frozen and trained parameters** across open omni embedders. *Inference* is the full parameter count at inference time; *Trained* is what training updates, and *Trained %* is trained over inference. For our 0.9B, the 68.32 M trained parameters are 65.37 M of projectors and pooler plus 2.95 M of $r=16$ LoRA adapters on the media encoders’ attention blocks; the adapters are merged into the encoder weights at export, so they add nothing at inference and the total stays at 935.35 M; Table 13 gives the per-component breakdown. The text backbone is never updated. gemini-embedding-2 is omitted because its parameter count is not disclosed.

audio. *Audio*. MS-CLAP (Elizalde et al., 2023) introduced contrastive language-audio pretraining and LAION-CLAP (Wu et al., 2023b) scaled it with feature fusion and keyword-to-caption augmentation; Whisper (Radford et al., 2023) (weakly supervised) and Dasheng (Dinkel et al., 2024) (self-supervised) are strong audio encoders without an aligned text tower.

Omni-modal embedders. LCO-Embedding-Omni-3B/7B (Xiao et al., 2026), BidirLM-Omni-2.5B (Boizard et al., 2026), NVIDIA’s omni-embed-nemotron-3B (Xu et al., 2025), and e5-omni-3B/7B (Chen et al., 2026) target a single space across text, image, audio and, apart from BidirLM-Omni, video, but are substantially larger than Omni-Embed-Mini-0.9B. Gemini Embedding 2 (Shanbhogue et al., 2026) is a closed, API-only native multimodal embedder whose parameter count is not disclosed; we report it for reference and exclude it from size-matched comparisons. The closest methodological contrast is e5-omni, which *calibrates* a LoRA-tuned shared space: modality-aware temperatures, a debiased negative curriculum, and batch whitening with a covariance-alignment loss together reduce mismatches between modalities. We instead anchor to a space that is already calibrated and never updated, so no calibration machinery is needed; our hybrid miner shares only their focus on hard negatives and false-negative suppression. e5-omni-7B leads us on the overall average (52.93 against 51.39), while our 2.3B edges ahead of e5-omni-3B (51.39 against 49.28). We lead on video (55.18 against 32.92 and 44.73) at a half to a quarter of their parameters, and hold a strong text score at 0.9B. Notably, in our text suite e5-omni

falls from 47.80 to 45.34 from 3B to 7B, which is consistent with the cost that a frozen anchor is designed to avoid. Crucially, absorbing every modality through joint contrastive updating puts the backbone’s native retrieval behaviour at risk, a critical concern for real-world deployment. Among the open omni embedders compared here, ours is the only one that leaves the backbone entirely frozen, text for the 0.9B variant and text plus vision for the 2.3B, adding new modalities through external modality encoders and projectors alone, so as to preserve the backbone’s native capability rather than degrade it (see Fig. 5).

Anchored vs. contrastive joint alignment. ImageBind (Girdhar et al., 2023) uses *images* as a binding modality: a frozen OpenCLIP ViT-H image encoder serves as anchor, and per-modality encoders are trained with symmetric InfoNCE to match the image embedding of naturally co-occurring pairs (image-audio from video, image-depth from RGB-D, etc.). Cross-modal alignment emerges because each modality is independently pulled toward the shared image space. LanguageBind (Zhu et al., 2024) adopts the same anchored-alignment structure but substitutes *language* for images as the binding modality, freezing a text encoder and LoRA-adapting per-modality encoders against it. AudioCLIP (Guzhov et al., 2022) takes a different joint-training approach, extending CLIP with a three-way contrastive loss across image, text, and audio. Omni-Embed-Mini shares LanguageBind’s choice of text as anchor but departs from both ImageBind and LanguageBind in four ways: (i) alignment adds *asymmetric distillation* (the frozen backbone embeds a dense caption as the teacher target) to the contrastive objective, so

each modality is also fitted to a fixed per-sample target vector rather than to batch-level contrast alone; (ii) pairing data is *manufactured* via dense captioning rather than requiring natural co-occurrence, removing ImageBind’s reliance on modality-specific paired corpora; LanguageBind also uses generated captions as contrastive text, whereas we additionally use them as a distillation target; (iii) the text encoder is not a separate frozen tower but the *identical backbone* that serves both teacher and student, so both are produced by the same untrained weights and text retrieval cannot regress; (iv) because that backbone is a single causal transformer, multiple modalities can be interleaved into one token sequence and encoded in a single pass, rather than each modality passing through its own tower before a late fusion step. Interleaved multi-modal queries are therefore native to the architecture rather than an added capability; we do not isolate their contribution experimentally, and leave a controlled study to future work. Conceptually, this is the cross-modal analogue of Reimers and Gurevych (2020), who aligned a new language to a frozen English teacher’s sentence space via distillation against translated sentences; we substitute “language” with “modality” and “translation” with “dense captioning.” Unlike their separately trained student, our text tower is frozen outright.

Our recipe builds on several established techniques. ANCE (Xiong et al., 2020) shows that periodically refreshing the hard-negative pool against the current encoder substantially improves

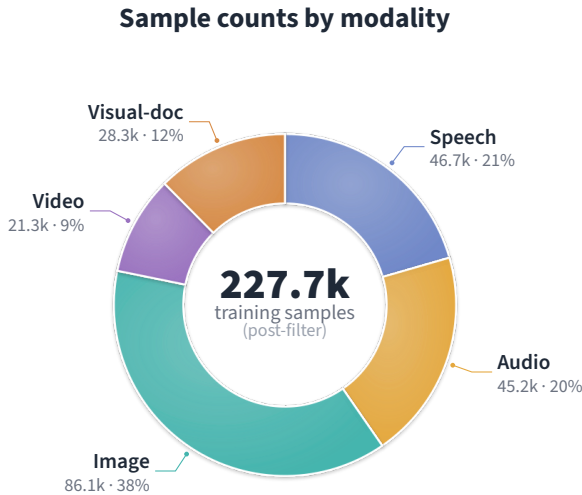


Figure 2: **Training data composition.** Sample-count distribution across the five non-text modalities in the caption-only slice of our omni-modal corpus, covering 227k effective rows after caption and duration filtering.

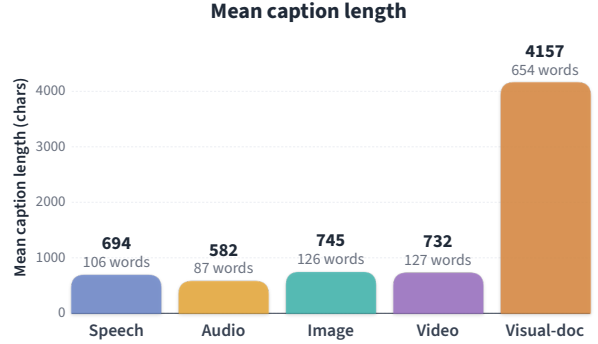


Figure 3: **Cascaded-caption length by modality.** Mean caption length per modality in the caption-only slice of our omni-modal corpus. Visual documents are $\sim 6\times$ longer because the expert-analyst captioner enumerates layout, headings, tables, and charts.

dense retrievers; our online hybrid miner is the multimodal-Matryoshka generalisation, with a text index and per-modality media FAISS-CPU (Johnson et al., 2019) indices refreshed on independent cadences. Matryoshka representation learning (Kusupati et al., 2022; OpenAI, 2024) enables a single trained embedding to be truncated post-hoc; we inherit MRL from the backbone and propagate it to all modalities through the distillation loss. SigLIP (Zhai et al., 2023) provides the pair-wise sigmoid contrastive loss, which scores each pair independently of the batch-level softmax normalisation and so performs well at small batch size, and LoRA (Hu et al., 2021) provides parameter-efficient adapters on the modality encoders.

3 Method

3.1 Asymmetric modality alignment

Two design principles and one training rule govern Omni-Embed-Mini. **(P1) Backbone invariance.** The backbone is never updated: no full fine-tuning, no LoRA on the backbone. This covers the text path for the 0.9B variant, and the text and vision paths for the 2.3B, which is built on a vision-language backbone. Those paths carry no projector and their weights are bit-identical to the stock backbone, so its calibrated retrieval quality is preserved by construction. **(P2) Cascaded-caption self-distillation.** Each media sample is paired with a dense caption; the embedding of that caption, obtained via the *same* frozen backbone, is the teacher target. Teacher and student share weights, so a cosine alignment loss in the native pooled hidden space suffices. **(T1) Alignment-first, then adaptation.** Phase 1 trains projectors only; Phase 2 (after

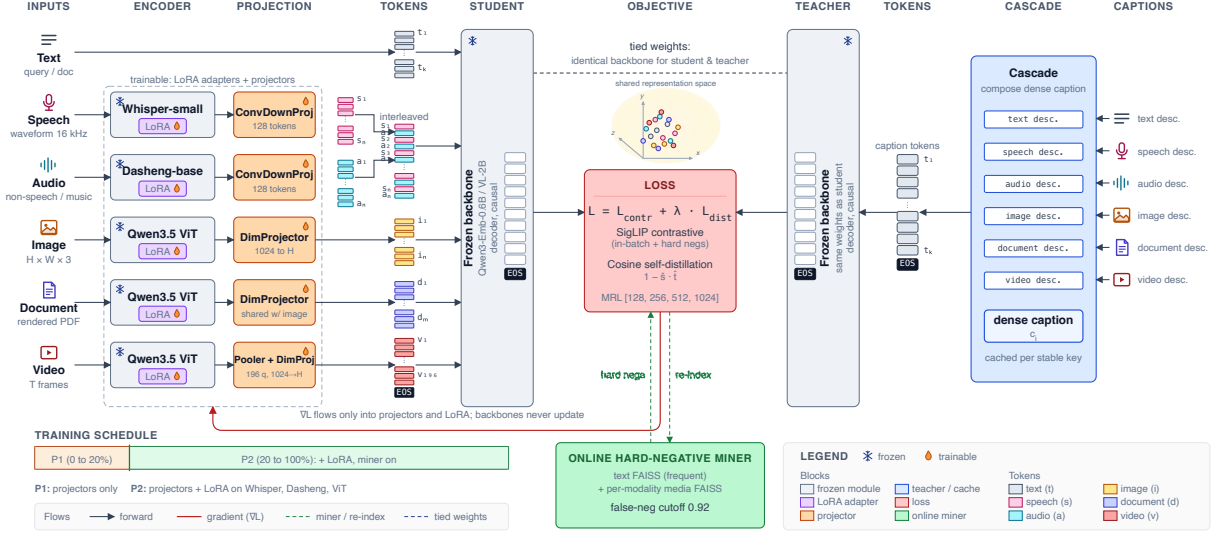


Figure 4: **Omni-Embed-Mini architecture.** Six student inputs flow through a single *frozen* causal text backbone; a cascaded-caption teacher mirrors them through the same backbone. Text passes through unmodified weights, so no text-side parameter is trained. Speech (Whisper) and audio (Dasheng) each emit 128 projected tokens, *temporally interleaved* into a 256-token column; image, document and video share a Qwen3.5 ViT, with video additionally spatially pooled to 196 tokens. The EOS-pooled hidden state is the embedding.

20% of steps) injects encoder LoRA. The motivation is that, before the projectors have aligned, encoder LoRA would adapt to noisy upstream projections; we adopt this schedule as a design choice and do not ablate the switch point.

3.2 Architecture

Fig. 4 gives the full data flow for both the student and the teacher path.

Backbone and pooling. A decoder-only transformer with causal attention; an EOS-pooling head extracts the last non-padding hidden state and L2-normalises it (no projection head). The 0.9B variant uses Qwen3-Embedding-0.6B (Zhang et al., 2025b) (hidden 1024) with a ViT extracted from Qwen3.5-0.8B (Qwen Team, 2026); the 2.3B variant uses Qwen3-VL-Embedding-2B (Li et al., 2026) (hidden 2048) with native vision.

Audio path: dual encoder, temporally interleaved. Whisper (Radford et al., 2023) (mel) and Dasheng (Dinkel et al., 2024) (raw waveform) run in parallel; each emits 128 projected tokens per 30 s chunk via a 1-D temporal down-projector with SwiGLU residual. The two streams are *interleaved in time*, $(w_1, d_1, w_2, d_2, \dots)$, yielding 256 tokens per chunk. Rather than a learned cross-attention bottleneck, temporal interleaving is intended as a structural inductive bias that encourages the backbone’s causal attention to attend jointly to semantic

speech content and acoustic texture within local context windows, at zero extra parameter cost.

Vision path. *Custom (0.9B):* ViT $\rightarrow 2 \times 2$ spatial merger $\rightarrow$ dimension projector $\rightarrow$ backbone hidden size; for video, a spatial pooler applies cross-attention against a learnable 196-query budget so video sequence length is independent of frame count. *Native (2.3B):* the backbone’s own visual module handles image and video; audio is injected via a forward hook on the embedding layer. Image and video tokens carry multimodal RoPE; audio tokens are spliced as a contiguous temporal span and reuse text RoPE. Media embeddings are spliced into modality-tagged placeholder positions via a one-hot matmul so gradients flow through the projectors and (Phase 2) encoder LoRA.

3.3 Cascaded-caption self-distillation

Each training sample i is paired with a dense caption c_i generated by Qwen3-Omni-30B-A3B-Instruct, with the source dataset’s ground-truth caption pinned in the system prompt as a grounding prior (§B). The teacher embedding t_i is the L2-normalised EOS-pooled output of the frozen backbone applied to c_i ; because the backbone is frozen, t_i is constant across training and is computed once and cached on disk under a stable sample id. The student embedding s_i is the same EOS-pool applied to the token sequence with media spliced in. Because s_i and t_i share the backbone and pool-

ing, they live in identical geometry; no auxiliary projection head is required.

3.4 Hybrid Matryoshka contrastive learning

Let $D = \{128, 256, 512, 1024, 2048\}$ denote the set of Matryoshka dimensions. For each $d \in D$, we truncate the embeddings to their first d coordinates and re-normalise them.

We compute a SigLIP pair-wise sigmoid loss (Zhai et al., 2023) over in-batch and hard-negative pairs, using a learned temperature τ and bias b :

$$\mathcal{L}_{\text{siglip}}^{(d)} = \mathcal{L}_{\text{siglip}}(\hat{s}^{(d)}, \hat{t}^{(d)}; \tau, b)$$

The Matryoshka contrastive loss aggregates the per-dimension losses with weights proportional to $1/\sqrt{d}$, thereby favouring aggressive truncations:

$$w_d = \frac{d^{-1/2}}{\sum_{d' \in D} d'^{-1/2}}, \quad \mathcal{L}_{\text{contr}} = \sum_{d \in D} w_d \mathcal{L}_{\text{siglip}}^{(d)}$$

We further use cosine self-distillation to align the student and teacher representations across Matryoshka dimensions:

$$\mathcal{L}_{\text{dist}} = \sum_{d \in D} w_d \mathbb{E}_i \left[1 - \hat{s}_i^{(d)} \cdot \hat{t}_i^{(d)} \right]$$

The final training objective is

$$\mathcal{L} = \mathcal{L}_{\text{contr}} + \lambda \mathcal{L}_{\text{dist}}, \quad \lambda = 1$$

Full equations are given in §G and implementation details in §H.

3.5 Online hybrid hard-negative mining

Hard negatives are supplied by a background subsystem maintaining FAISS-CPU (Johnson et al., 2019) inner-product indices (one for text, one per media modality), in the spirit of ANCE (Xiong et al., 2020). A lightweight *text* cycle ($\sim 10 \times / \text{epoch}$) re-embeds captions via the text-only path and retains the top- k neighbours per query. A more compute-intensive *media* cycle ($\sim 5 \times / \text{epoch}$) re-embeds samples per modality through the full model into per-modality indices. The miner shares the live model with the training loop, so each cycle returns negatives from the *current* representation: as the encoder improves, mined negatives become harder. Crucially, because the text branch never receives gradient, the text-side index is perfectly stable; only the media-side

index drifts, and it is refreshed so as to supply harder negatives as the encoder improves. This stability is intended to guard against *modality collapse* without ever destabilising text geometry. Concurrency, cache semantics, and failure handling are detailed in §D.

4 Training Data: Dense Caption Grounding

A central claim of this paper is that the success of frozen-backbone distillation depends critically on the *density* of the captioning signal. We therefore treat data preparation as a first-class methodological contribution rather than a logistical detail.

Source corpus. We train on the caption slice of our omni-modal caption corpus. From 590,858 total rows, filtering to caption-category rows across the speech, audio, image, video, and visual-doc configurations yields 242,080 single-turn samples; a subsequent duration filter (removing audio/video clips outside the supported length range) reduces this to 227,704 effective training rows (Fig. 2); caption length varies sharply by modality (Fig. 3). The omni split (cross-modal multi-turn chat) and chat samples are excluded; their narrow answers are unsuitable distillation targets.

Cascaded caption pipeline. Every row is the output of a single low-temperature ($T=0.7$) call to Qwen3-Omni-30B-A3B-Instruct, with the source dataset’s ground-truth caption interpolated as a “Reference caption” inside a modality-specific system prompt. This *grounds* the cascaded caption against the source, which may reduce unconstrained hallucination but does not preclude it. Visual-doc rows use an “expert document analyst” prompt with no reference caption (none exists). Full prompts and worked examples are in Appendix §B.

Dense captions are the *single most important data-side ingredient* of the recipe (as we demonstrate in §6.2): they expose more facets of each sample (objects, attributes, layout, temporal events, acoustic environment, OCR text) than short crowd-sourced captions, which provide a weaker, bag-of-words-like signal. We ablate this in Table 2, and worked qualitative examples per modality are in Appendix Table 10.

Because every teacher target is machine-generated, caption noise is a direct risk to the learned geometry. Four properties of the recipe

limit how far it can propagate: a grounded captioning prompt, a pooled-embedding target rather than token-level supervision, a 0.92 cosine cutoff in mining, and a frozen text path that no captioner bias can reach. These contain noise rather than verify its absence; we do not filter captions for factuality. §E gives the full argument and its limits.

5 Experiments

5.1 Benchmarks

Style	Sp.	Aud.	Img.	Vid.	Overall
Original Captions	41.75	31.02	17.63	4.74	23.78
Dense Captions	44.43	33.72	26.23	18.80	30.79
Δ	+2.68	+2.70	+8.60	+14.06	+7.01

Table 2: **Dense vs. original captions** on the 0.9B mining-ablation checkpoint (not the final checkpoint, whose scores appear in Fig. 5). *Original* denotes source-shipped captions, *Dense* our cascaded captions. Speech/audio are MAEB task means, image/video MMEB-V2 AvgImg/AvgVid, Overall their mean.

We evaluate on four community benchmarks; full schemas in Appendix §A. **MTEB-v2** (Muenighoff et al., 2023; Thakur et al., 2021): text retrieval, the 8-task BEIR English subset (ArguAna, CQA-E/P/Pr, FiQA, NFCorpus, SCIDOCS, SciFact); metric nDCG@10. Role: *text-side regression test*. **MAEB** (El Assadi et al., 2026): 22 English audio tasks (retrieval, classification, clustering, multi-label, pair, reranking, zero-shot); mixed metrics. Role: cross-modal audio-to-text alignment plus audio coverage. **MMEB-V2** (Meng et al., 2025): 16 English tasks (10 image, 6 video); metric hit@1. **ViDoRe-V3** (Loison et al., 2026): 7 English visually-rich domains; metric nDCG@10.

English-only protocol. MMEB, MAEB, and ViDoRe are taken almost in full (16/18, 22/22, 7/7 supported English tasks); MTEB is reduced to BEIR-8 because the full v2 catalog has 1000+ task-language pairs. Each benchmark is re-validated by re-running its published anchor in our harness; full protocols are in §A. Three training sources share a parent corpus with MMEB-V2 tasks; an identifier-level overlap audit finds zero shared items in every case (§C.1).

5.2 Setup

Hardware. 8× AMD Instinct MI210 (64 GB HBM2e each), bf16, DDP.

Hyperparameters and cost. AdamW (wd 0.01, clip 1.0); 2% linear warm-up → cosine; LR 2×10^{-4} for audio projectors, 1×10^{-4} for vision projectors and encoder LoRA (Phase 2 only); LoRA $r=16$, $\alpha=32$; phase switch at 20%; MRL dims {128, 256, 512, 1024} (+2048 for 2.3B); $\lambda_{\text{dist}} = 1$. Mining ablations are run in a two-epoch setting. Full table in §H. For video evaluation on MMEB-V2, all omni-style models embed 8 frames as images and mean-pool. The 8 frames are the benchmark’s own pre-extracted frames rather than a sampling choice of ours, so this keeps comparisons uniform, but it caps achievable video performance (§I).

5.3 Main results

Fig. 5 summarises the headline result restricted to the omni class. At **0.9B**, our model is the smallest open omni embedder by $\sim 2.7 \times$ relative to the next entry (BidirLM-Omni-2.5B) and $\sim 9.5 \times$ relative to LCO-Embedding-Omni-7B, and it is the only one among the compared omni embedders whose text branch is untrained (§6.1). At 49.57 on *Text* it is second among open omni embedders only to omni-embed-nemotron-3B (50.48), which is five times larger. On *Speech* and *Audio* the 0.9B trails the larger omni baselines, the expected cost of adding these modalities through small projectors rather than by retraining the backbone. The **2.3B** variant, built on a frozen Qwen3-VL-Embedding-2B, reaches an overall-modality average of 51.39: it leads the omni block on *Video* (55.18) and is competitive on *Image* (64.80) and *Visual-Doc* (58.10), where e5-omni-7B, at $3.7 \times$ its size, is the strongest open omni model. Against gemini-embedding-2, which is closed and of undisclosed size, our 2.3B leads on the overall-modality average (51.39 to 49.51). The all-class version of these panels (text-only, VL, audio/CLAP, and omni baselines together) is in Appendix Fig. 7; per-task tables are in Appendix §A.

The 0.9B variant scores lower on *Image* (26.29) and *Video* (18.48) because its vision encoder is extracted from a 0.8B backbone with a comparatively small ViT, trained on only $\sim 86\text{K}$ image and $\sim 36\text{K}$ video rows. The 2.3B variant, with the native Qwen3-VL vision encoder, closes this gap (64.80 image, 55.18 video), which is consistent with vision-encoder capacity, rather than the alignment recipe, limiting the 0.9B.

Routing every modality through a shared space also retains within-modality retrieval without a



Figure 5: **Per-modality rankings, omni-only.** Same panels as Appendix Fig. 7, restricted to omni embedders. The 2.3B variant leads the omni block on video and is competitive on image and visual documents; speech and audio remain our weakest modalities. gemini-embedding-2 is closed and shown for reference only.

dedicated training signal. On the image-to-image tasks NIGHTS and FashionIQ (Table 7), the 2.3B scores 68.00 and 34.50, in line with its frozen Qwen3-VL-Embedding-2B backbone (67.80 and 37.50), and the 0.9B, which is never trained on an image-to-image objective, keeps much of its raw vision encoder’s behaviour (53.90 against 59.30 on NIGHTS, 7.40 against 3.80 on FashionIQ). See Appendix §F.

6 Ablations and Analysis

Targeted ablations isolate four design choices: frozen backbone, dense captions, hybrid mining, and Matryoshka propagation.

6.1 Backbone invariance

On MTEB-v2 BEIR-8, Omni-Embed-Mini-0.9B scores **49.57**. This is not a tuning artefact: the text-only inference path touches only the unchanged backbone weights, never the trained projectors or LoRA adapters, and those weights are bit-identical to the pretrained Qwen3-Embedding-0.6B. Training consequently cannot move this score, and independently trained checkpoints agree on all eight tasks to four decimal places. Per-task numbers are in Appendix Table 4. The 2.3B variant inherits

Modality	0.9B			2.3B		
	Frozen	+LoRA	Δ	Frozen	+LoRA	Δ
Text	49.57	46.44	−3.13	47.94	47.29	−0.65
Speech	44.43	47.09	+2.67	41.55	31.02	−10.53
Audio	33.72	34.83	+1.10	33.97	25.57	−8.40
Image	26.23	30.04	+3.81	64.80	64.34	−0.46
Video	18.80	24.46	+5.66	55.18	54.60	−0.58
Vis-Doc	46.98	51.04	+4.06	58.10	58.02	−0.08

Table 3: **Frozen backbone against backbone LoRA.** Both columns are ablation checkpoints, not the released ones. At 0.9B, unfreezing the backbone improves every media modality and costs text; at 2.3B it degrades every modality, and far more on the speech and audio paths that Phase 2 trains than on the untrained text, image and video paths.

the same guarantee over a wider surface: both its text and its vision paths are left untrained, so its image and video behaviour is inherited from the vision-language backbone.

Unfreezing the backbone. Adding LoRA to the backbone itself behaves differently at the two scales (Table 3). At 0.9B it is a genuine trade-off: every media modality improves, by +1.10 to +5.66, while text drops by 3.13 points, which is exactly the regression the frozen anchor exists to prevent. At

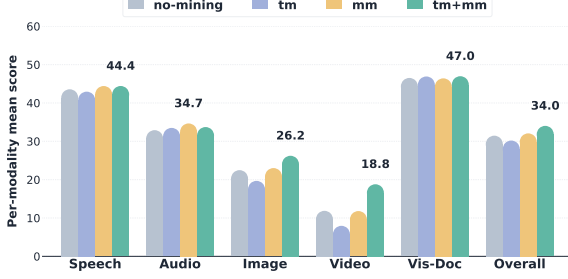


Figure 6: **Mining ablation.** Per-modality means on the 0.9B recipe across no-mining (in-batch only), text-only, media-only and combined mining; text is invariant by construction. Combined mining gives the best overall average and the largest image and video gains; visual documents are near-insensitive (all checkpoints within 0.6 nDCG@10).

2.3B the same intervention degrades every modality, most sharply the two the adapters were trained for (speech -10.53 , audio -8.40), while text, image and video shift by under a point; here backbone updates appear to disturb the alignment that Phase 2 establishes rather than add capacity. Text improved at neither scale. We therefore keep the backbone frozen, which preserves text retrieval at both scales and gives the new modalities a stable target space.

6.2 Dense vs. original captions

Table 2 shows the smallest gap is on speech ($+2.68$), where source transcripts already carry most of the content; the largest is on video ($+14.06$), where short clip captions miss nearly every temporal event. The same effect motivated the dense recaptioning pipeline of Li et al. (2024) for VLM pre-training; it is equally decisive for retrieval.

6.3 Hybrid mining

Fig. 6 shows that only combined mining separates clearly from no mining ($+2.6$ on the mean over the five media modalities); the gaps among the other three configurations are comparable to the seed spread reported below. Text-only mining alone slightly hurts on several modalities (the student sees harder text negatives but no harder media ones, biasing the projector); media-only gives a small improvement on most modalities; combining both yields the best overall score, though per-modality rankings swap on video and on visual documents, where media-only falls marginally below no-mining. Visual-document retrieval is in fact the modality least sensitive to mining: the four configurations agree to within 0.6 nDCG@10

on ViDoRe-V3 (46.42 to 46.98). What moves that modality is caption quality rather than negative sampling, as replacing the dense cascaded captions with the datasets’ original captions costs 4.8 nDCG@10 (46.98 to 42.21). The two miners contribute complementary pressure: text-mining sharpens the caption-text geometry the teacher signal targets, while media-mining sharpens the modality-specific student geometry; dropping either leaves a slack axis along which the projector can drift. The main recipe uses both miners.

Seed sensitivity. Across three random seeds (42, 71 and 1234) of the 10-epoch no-mining configuration, the model is stable across all five trained modalities: image scores 27.48 ± 0.43 , general audio 33.96 ± 0.59 , video 19.82 ± 0.95 , speech 43.10 ± 1.03 , and visual-document retrieval 45.48 ± 1.76 (mean $\pm$ standard deviation over the three seeds). Text is identical across all three at 49.57: the text branch carries no trainable parameters, so every 0.9B checkpoint returns the same value. Per-seed numbers are in §F.

6.4 Matryoshka behaviour

Because MRL slicing acts on the backbone’s pooled hidden state with no learned head, Matryoshka behaviour at every dimension is inherited from the backbone (Kusupati et al., 2022; OpenAI, 2024). Averaged-probe AVG falls only slightly from 1024 to 128 for both variants, whereas the omni-specific baselines fall away far more steeply (Fig. 8). For the 0.9B, a 128-dim truncation reduces on-disk footprint $8\times$: a 10M-item omni index in float32 fits in ~ 5 GB rather than ~ 41 GB, at modest cost in quality. Further analyses are in §F.

7 Conclusion

We introduced **Omni-Embed-Mini**, a 0.9B omni-modal retrieval model that binds text, speech, audio, images, video, and visually-rich documents into a unified space by recasting cross-modal alignment as self-distillation through a shared, *frozen* text backbone. By keeping the text encoder fixed and aligning every other modality to its representation, we sidestep the catastrophic forgetting that typically plagues multi-modal fine-tuning and turn omni-modal training into a lightweight projector-plus-LoRA exercise. The model preserves its backbone’s text retrieval quality while remaining competitive with open omni embedders $\sim 2.7\times$ to $9.5\times$ larger, and carries over to a 2.3B variant competi-

tive with baselines $2\times$ to $4\times$ its size. Because the backbone is frozen, future text-embedder improvements can in principle be adopted by re-running only the projector and LoRA training, as our backbone swap from the 0.6B text encoder to Qwen3-VL-Embedding-2B illustrates. Open threads include learned mixing of the dual audio-encoder tokens, a Mixture-of-Encoders variant that routes inputs to specialist branches, and replacing per-frame mean-pooling for video (§I) with a native spatial pooler that better preserves temporal and spatial structure. Beyond architecture, the most pressing community gap is evaluation: existing benchmarks each probe a single modality slice, and none evaluate cross-modal retrieval in arbitrary query→corpus combinations across all six modalities simultaneously.

Limitations

Omni-Embed-Mini is a research prototype for omni-modal retrieval and is not validated for high-stakes deployment. Although the model provides efficient alignment across text, speech, audio, image, video, and visually-rich documents, it may still retrieve incorrect, biased, private, or misleading items when used in downstream systems. Therefore, it should not be used as a standalone decision system in sensitive domains without task-specific validation and human oversight.

The model’s shared embedding space is constrained by the frozen text backbone used to preserve text-side retrieval performance. As a result, modality-specific phenomena that are difficult to verbalise, such as complex acoustic textures, abstract visual patterns, or fine-grained temporal cues, may not be fully represented. The cascaded-caption self-distillation pipeline also depends on teacher-generated captions, which may contain hallucinations, omissions, or biases that are inherited by the student model. We do not filter captions for factuality, and we do not verify them against their source media; the only automatic check in the pipeline is a 0.92 cosine threshold used to discard likely false negatives during mining. We also did not run a controlled caption-perturbation study, so we cannot quantify how far retrieval quality degrades as caption quality falls. What we can report is that caption quality matters a great deal in aggregate: replacing our dense cascaded captions with the datasets’ original captions costs 7.01 points on the mean of speech, audio, image and video (Table 2)

and 4.8 nDCG@10 on visual documents. Because a single captioner produces every target, any systematic bias it carries is applied uniformly across the corpus rather than averaged out, which is the failure mode we would most want to measure. Plausible mitigations we have not implemented include cross-checking captions against a second captioner, filtering on caption-to-media agreement scored by an independent model, and sampling captions for human verification; we regard a caption-robustness study along these lines as the most useful next step for this recipe.

The audio pathway currently uses a fixed temporal interleaving of speech and environmental-audio tokens, which may be suboptimal for inputs dominated by only one audio type. In addition, the compact 0.9B and 2.3B model scales are optimised for efficiency, but may limit performance on complex reasoning-heavy multimodal retrieval tasks.

Finally, the evaluation is limited by the coverage of available benchmarks. Existing datasets do not yet provide a unified any-to-any retrieval benchmark across all six supported modalities, so our results may not generalise to all query-corpus combinations, languages, domains, data qualities, or real-world retrieval workflows.

Ethical Considerations

This work is intended for research and educational use in omni-modal representation learning and retrieval. Omni-Embed-Mini should not be used for autonomous decision-making in high-stakes domains such as medicine, law, finance, hiring, education, security, or public-sector services without appropriate validation and human supervision.

The main ethical risks include inaccurate retrieval, bias amplification, privacy leakage, and misuse. Since the model aligns multiple modalities into a shared embedding space, errors or biases from training data, teacher-generated captions, and pre-trained encoders may affect retrieval quality across languages, accents, visual appearances, environments, document types, and cultural contexts. Downstream users should conduct safety, fairness, privacy, robustness, and human-in-the-loop evaluations before applying the model in sensitive settings.

We also disclose our use of generative AI tools. Writing-support models were used for grammar, style, and clarity improvements. All technical claims, experimental results, analyses, and con-

clusions were authored, checked, and verified by the authors; any coding assistance was limited to boilerplate scaffolding, minor refactoring, or formatting support.

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Appendix

This Appendix section expands on the main paper with full benchmark protocols and per-task results for all four benchmarks (§A), the cascaded-caption data-generation pipeline with system prompts and one qualitative example per modality (§B), per-source dataset attributions and licenses together with the train/test overlap audit (§C), the mining subsystem (§D), caption grounding and noise safeguards (§E), additional ablations covering Matryoshka truncation, per-seed results and emergent media-to-media alignment (§F), the full loss equations (§G), hyperparameters and the per-component parameter breakdown (§H), and additional limitations (§I).

A Detailed benchmark protocols

For each benchmark we briefly describe the source benchmark, the exact subset of tasks we evaluate, and the metric. We then report per-task numbers for the established public baselines and our Omni-Embed-Mini variants. Parameter counts in the left-most numeric column are taken from each model’s published card; closed-source models with undisclosed sizes are marked “?”.

Baselines. We compare against the following public models. *Text*: Qwen3-Embedding (Zhang et al., 2025b), voyage-3-m-exp (Voyage AI, 2025), Conan-embedding-v2 (TencentBAC, 2025), inf-retriever (Yang et al., 2025), LENS (Lei et al., 2025), GritLM (Muennighoff et al., 2025), Jasper (Zhang et al., 2025a), embeddinggemma (Vera et al., 2025) and bge-large-en (Xiao et al., 2024). *Vision-language*: Qwen3-VL-Embedding (Li et al., 2026), VLM2Vec-V2 (Meng et al., 2025), RzenEmbed (Jian et al., 2025), Ops-MM-embedding (OpenSearch-AI Team, Alibaba Cloud, 2025), seed-1.6-embedding (ByteDance Seed, 2025), WeMM-Embedding (Zhou et al., 2026), IFM-TTE (Cui et al., 2026), colqwen3.5 (athrael soju, 2026), nomic-embed-multimodal (Team, 2025), llama-nemotron-embed-v1 (NVIDIA, 2025), SigLIP (Zhai et al., 2023) and the vision encoder of Qwen3.5-0.8B (Qwen Team, 2026). *Audio*: MS-CLAP 2022 (Elizalde et al., 2023) and 2023 (Elizalde et al., 2024), LAION-CLAP (Wu et al., 2023b), Qwen2-Audio (Chu et al., 2024), and the Whisper (Radford et al., 2023) and Dasheng (Dinkel et al., 2024) encoders.

Omni-modal: LCO-Embedding-Omni (Xiao et al., 2026), BidirLM-Omni (Boizard et al., 2026), omni-embed-nemotron (Xu et al., 2025), e5-omni (Chen et al., 2026) and the closed gemini-embedding-2 (Shanbhogue et al., 2026).

Table conventions. In the per-benchmark tables below, the `omni` block marks open omni embedders. Within it, per-column best is in **bold** and second best underlined, computed over the open omni embedders and our two `final checkpoints`; the closed gemini-embedding-2, marked [‡], and the `mining ablation checkpoints` are excluded from that comparison. gemini-embedding-2 is API-only with an undisclosed parameter count, so it is reported for reference and excluded from size-matched comparisons. Indented rows are the hard-negative-mining ablation checkpoints; the two `highlighted` rows are our final checkpoints, trained for longer under the best ablation configuration (text and media mining with dense captions), and are the models reported in the main text and in Fig. 5.

A.1 MTEB-v2 (8-task English BEIR subset)

What it is. MTEB-v2 (Muennighoff et al., 2023) is the second release of the Massive Text Embedding Benchmark; it modernises the evaluation API and unifies multilingual (MMTEB) and image (MIEB) extensions, with the full v2 catalog spanning 1000+ task-language pairs across 250+ languages.

Subset evaluated. Even at sub-billion parameters, evaluating the full MTEB-v2 catalog requires days of GPU time, and many embedding papers report on a curated BEIR subset for the same reason. We use an 8-task English BEIR subset spanning short and long queries against short and long corpora: ArguAna, CQADupstack-English, CQADupstack-Physics, CQADupstack-Programmers, FiQA-2018, NFCorpus, SCIDOCS, SciFact.

Metric. nDCG@10 ($\times 100$) per task; Avg is the mean across the 8 tasks. Per-task results are reported in Table 4.

A.2 MAEB (22-task English subset)

What it is. MAEB (El Assadi et al., 2026) is a unified MTEB-ecosystem benchmark for audio embeddings that covers speech, environmental sound, and music with both audio-only and text-query tasks.

Subset evaluated. We evaluate a 22-task subset comprising the 15 native-English tasks plus 7

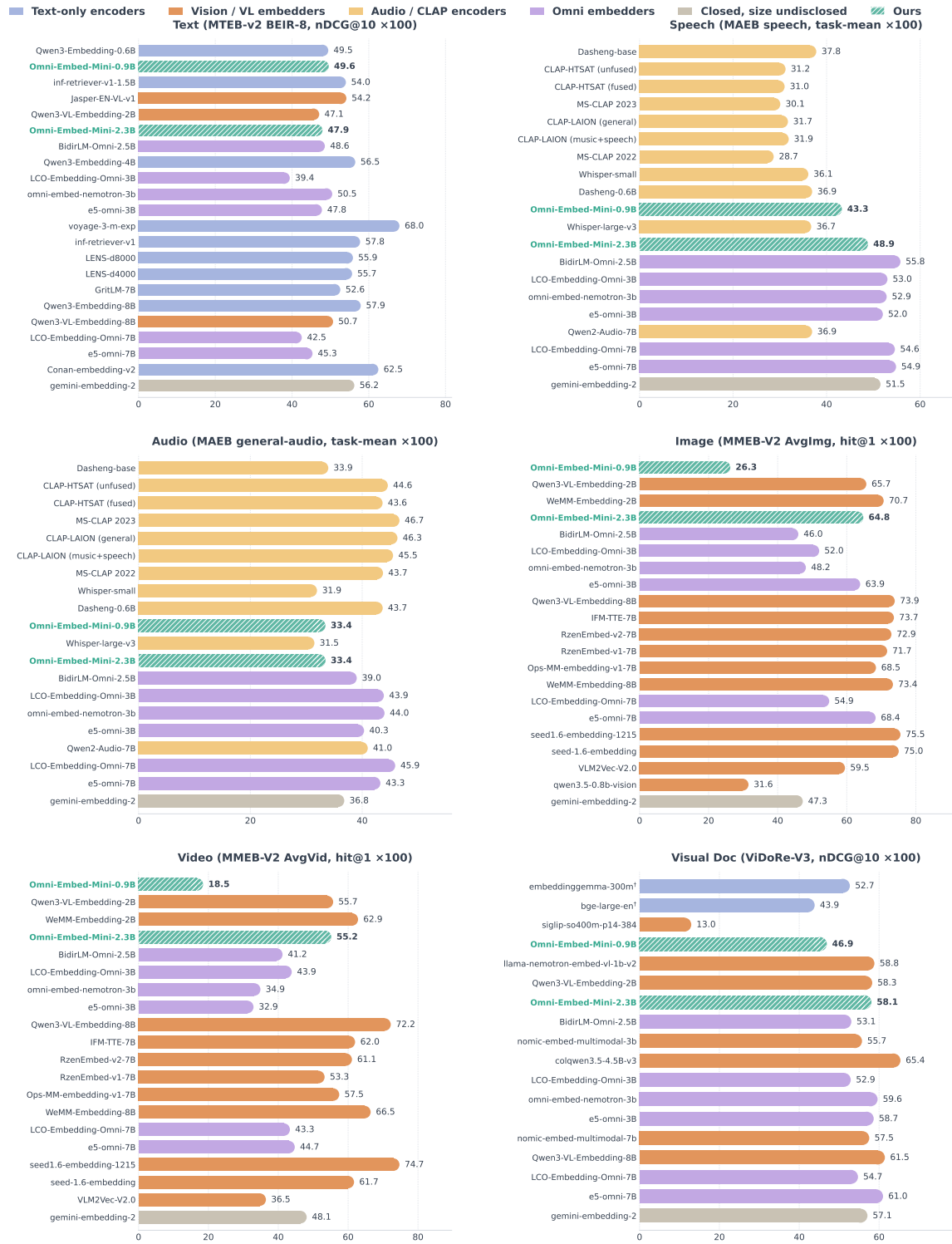


Figure 7: **Per-modality rankings, all classes.** Each panel shows every model with a score for that modality, ordered by inference parameter count, smallest first; bars are coloured by class (text-only, VL/vision, audio/CLAP, omni, ours). gemini-embedding-2 is closed and of undisclosed size, so it is drawn in its own colour, placed last, and shown for reference only. [†]In the Visual Doc panel, embeddinggemma-300m and bge-large-en are text retrievers scored on the OCR’d page text alone; every other model there is scored under the image+text protocol and sees the page image, so those two bars are not a like-for-like comparison. Text is MTEB-v2 BEIR-8 Avg; Speech/Audio are mean across the 12 speech and 10 general-audio MAEB tasks; Image/Video are MMEB-V2 AvgImg/AvgVid; Visual Doc is ViDoRe-V3 Avg nDCG@10. Omni-Embed-Mini-0.9B reaches 49.57 MTEB nDCG@10 with text weights bit-identical to the backbone, so training cannot regress text retrieval, while extending to five additional modalities at $\sim 2.7\times$ to $9.5\times$ fewer parameters than every open omni baseline. The 2.3B variant additionally inherits Qwen3-VL-Embedding-2B image/video/visual-doc strengths.

Model	Params	ArgA	CQA-E	CQA-P	CQA-Pr	FiQA	NFC	SCID	SciF	Avg
<i>Text embedders</i>										
voyage-3-m-exp	6.9B	89.79	69.98	73.71	65.87	78.01	<u>46.99</u>	34.53	<u>85.09</u>	68.00
Conan-embedding-v2	1.5B	77.79	<u>57.50</u>	<u>60.58</u>	<u>60.66</u>	<u>71.86</u>	55.74	31.63	84.27	<u>62.50</u>
Qwen3-Embedding-8B	7.6B	76.85	57.44	58.31	53.49	64.57	41.45	<u>32.74</u>	78.46	57.91
inf-retriever-v1	7.1B	<u>84.86</u>	55.96	52.35	46.64	62.35	43.66	30.78	85.40	57.75
Qwen3-Embedding-4B	4.0B	75.64	55.00	56.58	51.17	62.65	41.10	31.44	78.33	56.49
LENS-d8000	7.1B	76.02	55.47	53.89	51.15	61.57	40.61	28.54	79.75	55.87
LENS-d4000	7.1B	77.32	55.62	52.87	51.53	60.43	41.64	27.51	78.39	55.66
inf-retriever-v1-1.5B	1.5B	81.50	52.16	49.68	45.19	56.09	38.55	26.34	82.84	54.04
GritLM-7B	7.2B	63.17	52.10	52.85	48.29	59.91	40.86	24.41	79.13	52.59
Qwen3-Embedding-0.6B	596M	69.45	51.46	50.69	46.28	47.32	36.35	24.29	69.76	49.45
<i>Vision-language embedders</i>										
Jasper-EN-VL-v1	1.6B	65.20	54.27	55.00	50.45	60.12	43.31	24.64	80.37	54.17
Qwen3-VL-Embedding-8B	8.1B	62.77	52.88	52.13	46.50	47.62	39.75	24.72	79.19	50.70
Qwen3-VL-Embedding-2B	2.1B	58.46	48.27	50.66	43.44	42.30	37.05	21.79	74.73	47.09
<i>Omni / audio-language embedders</i>										
BidirLM-Omni-2.5B	2.5B	57.04	49.91	51.67	45.93	45.60	<u>38.67</u>	23.32	76.31	48.56
LCO-Embedding-Omni-7B	8.9B	39.09	44.47	50.59	43.18	39.21	36.35	19.97	67.42	42.54
LCO-Embedding-Omni-3B	4.7B	41.39	39.51	45.67	40.14	35.47	31.86	18.21	63.13	39.42
omni-embed-nemotron-3B	4.7B	59.00	55.45	54.64	49.57	53.16	36.31	21.70	74.01	50.48
e5-omni-3B	4.7B	52.60	46.29	50.15	41.90	51.24	37.92	24.46	77.88	47.80
e5-omni-7B	8.9B	40.17	43.20	47.81	39.91	<u>51.29</u>	40.44	22.74	<u>77.13</u>	45.34
gemini-embedding-2 [‡]	?	71.61	58.16	56.47	51.62	53.44	43.74	24.82	90.10	56.24
Omni-Embed-Mini-0.9B	935M	69.54	<u>51.62</u>	50.77	<u>46.17</u>	47.47	36.83	<u>24.38</u>	69.80	<u>49.57</u>
Omni-Embed-Mini-2.3B	2.3B	<u>59.72</u>	48.98	<u>52.74</u>	44.98	43.79	37.00	22.04	74.26	47.94

Table 4: **MTEB-v2 BEIR-8 per-task scores.** All eight tasks measure text retrieval ranking quality, scored by nDCG@10 ($\times 100$). **Task abbreviations:** ArguAna (ArgA), CQADupstack-English (CQA-E), CQADupstack-Physics (CQA-P), CQADupstack-Programmers (CQA-Pr), FiQA-2018 (FiQA), NFCorpus (NFC), SCIDOCs (SCID), SciFact (SciF). Bolding, colour and footnote conventions follow §A. Omni-Embed-Mini-0.9B applies no LoRA or projector to its text branch, whose weights are bit-identical to the frozen Qwen3-Embedding-0.6B backbone, so training cannot regress text retrieval.

language-independent music/environmental-sound tasks; multilingual source tasks are restricted to their English split. Tasks split naturally into a 10-task *audio* subset and a 12-task *speech* subset, each reported in a separate table below. The benchmark task families, following (El Assadi et al., 2026), are: text-to-audio retrieval (T2A), classification (Cls), clustering (Clu), multi-label tagging (Mlt), pair-classification (Pair), reranking (Rank) and zero-shot classification (ZSh).

Metrics. T2A reports HitRate@5; Cls and ZSh report accuracy; Clu reports V-measure; Mlt reports LRAP; Pair reports max average precision; Rank reports MAP@1000. Avg is the mean across each model’s scored tasks within the relevant modality subset. Per-task results are reported in Table 5 (audio) and Table 6 (speech).

A.3 MMEB-V2 (16-task English subset)

What it is. MMEB-V2 (Meng et al., 2025) is a multimodal embedding benchmark combining image and video tasks under a unified hit@1 retrieval and classification protocol.

Subset evaluated. We use the 16 English tasks: 10 image tasks (VOC2007, Country211, OK-VQA, EDIS, MSCOCO-T2I, VisualNews-T2I, MSCOCO-I2T, VisualNews-I2T, NIGHTS, FashionIQ) and 6 video tasks (HMDB51, UCF101, MSR-VTT, MSVD, DiDeMo, VATEX). Following the benchmark task taxonomy, image tasks span zero-shot image classification (CLS), visual question answering (VQA), text-to-image (T2I), image-to-text (I2T) and image-to-image (I2I) retrieval; video tasks span zero-shot action classification (V-CLS) and text-to-video (V-T2V) retrieval. The

Model	Params	Clotho HR@5	MACS HR@5	US8K HR@5	BjOp. Acc	Bird Acc	GTZAN Acc	Mrd. Acc	Veh. V-meas.	FSD19 LRAP	GTZ-R MAP@1k	Avg
<i>Audio-language models</i>												
Qwen2-Audio-7B	7.0B	0.45	1.53	0.22	97.45	37.10	93.10	61.17	5.52	32.64	80.85	41.00
<i>CLAP-style dual encoders</i>												
CLAP-LAION (general)	194M	33.72	33.08	0.96	93.62	17.00	84.50	49.45	3.37	80.63	66.78	46.31
CLAP-LAION (music+speech)	194M	32.39	30.53	0.94	91.52	16.40	83.50	48.99	4.72	80.62	65.65	45.53
MS-CLAP 2023	160M	41.93	27.23	0.94	91.11	17.30	78.10	52.09	2.86	79.73	75.43	46.67
MS-CLAP 2022	196M	58.73	41.73	0.88	93.20	13.20	58.70	29.14	7.98	70.86	62.91	43.73
CLAP-HTSAT (unfused)	153M	34.18	27.99	0.98	91.96	16.40	74.90	49.52	2.66	80.84	66.45	44.59
CLAP-HTSAT (fused)	154M	33.57	29.26	0.94	92.77	18.00	67.90	47.00	4.71	81.09	61.30	43.65
<i>Speech encoders (no text tower; avg over 7 non-retrieval columns[†])</i>												
Dasheng-0.6B	630M	–	–	–	84.33	8.50	76.20	53.83	5.76	7.09	69.98	43.67 [†]
Dasheng-base	86M	–	–	–	76.73	5.10	65.40	44.50	3.22	7.08	35.56	33.94 [†]
Whisper-large-v3	1.5B	–	–	–	64.80	5.50	45.90	30.80	1.49	7.99	63.77	31.46 [†]
Whisper-small	244M	–	–	–	67.38	7.30	47.40	27.20	0.98	9.05	64.16	31.92 [†]
<i>Omni embedders</i>												
LCO-Embedding-Omni-7B	8.9B	33.89	29.77	<u>0.92</u>	<u>92.79</u>	34.10	82.30	<u>42.07</u>	3.63	<u>60.96</u>	78.71	45.91
LCO-Embedding-Omni-3B	4.7B	32.70	22.65	0.88	93.22	31.60	81.00	39.54	2.18	59.67	75.37	43.88
BidirLM-Omni-2.5B	2.5B	12.05	8.40	0.65	90.25	<u>33.10</u>	74.10	45.35	5.11	46.52	74.76	39.03
omni-embed-nemotron-3B	4.7B	33.25	25.19	0.88	88.60	28.60	<u>82.50</u>	32.69	<u>4.56</u>	61.60	82.12	<u>44.00</u>
e5-omni-3B	4.7B	<u>39.48</u>	<u>27.48</u>	0.84	76.71	24.00	78.30	30.17	3.71	37.76	<u>84.97</u>	40.34
e5-omni-7B	8.9B	42.08	26.72	0.96	81.81	24.50	84.60	30.66	2.48	53.14	85.67	43.26
gemini-embedding-2 [†]	?	15.36	6.62	0.67	84.35	24.10	76.80	36.76	6.32	36.91	80.07	36.80
<i>Ours: Omni-Embed-Mini</i>												
0.9B, no mining	935M	13.38	6.87	0.75	79.66	15.70	66.50	28.15	2.90	39.41	75.75	32.91
0.9B, text mining	935M	8.16	3.82	0.67	85.60	18.40	66.10	37.22	8.19	35.30	71.56	33.50
0.9B, media mining	935M	13.29	6.87	0.65	83.46	18.40	69.00	35.26	3.82	39.18	76.66	34.66
0.9B, text + media	935M	12.62	7.89	0.16	81.37	14.30	66.80	30.50	8.80	39.03	75.77	33.72
0.9B, t+m, orig. captions	935M	18.89	10.69	0.69	74.57	12.90	50.30	37.15	6.24	34.13	64.60	31.02
2.3B, text + media	2.3B	26.57	9.16	0.86	77.54	16.50	66.80	26.17	1.86	39.11	75.13	33.97
Omni-Embed-Mini-0.9B	935M	9.24	6.11	0.65	83.88	16.20	66.30	34.40	2.58	38.22	76.72	33.43
Omni-Embed-Mini-2.3B	2.3B	17.12	8.91	0.41	84.29	18.70	65.30	32.59	0.75	34.77	71.57	33.44

Table 5: MAEB-Audio per-task scores across 10 tasks. **Task abbreviations:** Clotho-T2A (Clotho), MACS-T2A (MACS), UrbanSound8K-T2A (US8K), BeijingOpera (BjOp.), BirdCLEF (Bird), GTZAN-Genre (GTZAN), MridinghamTonic (Mrd.), VehicleSoundClustering (Veh.), FSD2019Kaggle (FSD19), GTZAN-AudioReranking (GTZ-R). **Metrics** appear below each task name: HR@5 = HitRate@5, Acc = accuracy, V-meas. = V-measure, LRAP = label ranking avg. precision, MAP@1k = mean avg. precision at 1000. Bolding, colour and footnote conventions follow §A. [†]Speech-only encoders lack a text tower and are not scored on the three retrieval tasks; their average is over the remaining seven columns.

MMEB-V2 NExTQA and EgoSchema video-QA tasks are not included in this subset.

Metric. hit@1 ($\times 100$) per task; AvgImg, AvgVid and AvgAll are means across the image, video, and combined task sets respectively. Per-task results are reported in Table 7 (image tasks) and Table 8 (video tasks).

A.4 ViDoRe-V3 (7 English domains)

What it is. ViDoRe-V3 (Loison et al., 2026) is a visual-document retrieval benchmark over visually rich enterprise documents (financial reports, HR handbooks, scientific papers, regulatory filings, pharmaceutical and industrial manuals): given a text query, retrieve the relevant page-image.

Subset evaluated. We evaluate the 7 English-source domains (Finance, HR, Industrial, Pharmaceuticals, Computer Science, Energy, Physics); the French-only Finance task is excluded.

Metric. nDCG@10 per task; Avg is the mean across the 7 domains. The public ViDoRe leaderboard headline reports nDCG@5; we report nDCG@10 to match the convention used elsewhere in this paper. Per-domain results are reported in Table 9.

B Cascaded-caption data generation

The caption slice of our omni-modal corpus is the sole training corpus for the contrastive objective. Every row is the output of a single multimodal

Model	Params	GigaSp HR@5	SpSQuAD HR@5	CREMA Acc	AgeDet Acc	IEMOCAP Acc	VoxCel Acc	CREMA-C V-meas.	CREMA-P max-AP	NMSQA max-AP	VoxPop max-AP	Ravd ZS Acc	SpCmd ZS Acc	Avg
Audio-language models														
Qwen2-Audio-7B	7.0B	0.09	1.00	73.99	17.59	92.96	29.54	32.37	68.87	48.85	52.93	14.37	10.38	36.91
CLAP-style dual encoders														
CLAP-LAION (general)	194M	0.16	2.00	39.83	20.51	89.28	30.47	13.17	54.71	47.42	53.55	17.29	12.44	31.74
CLAP-LAION (music+speech)	194M	0.21	2.00	40.18	16.59	93.12	33.78	14.43	56.66	46.14	53.49	17.29	9.03	31.91
MS-CLAP 2023	160M	0.10	0.00	37.06	15.35	85.86	26.18	10.73	57.65	51.09	52.05	15.21	10.01	30.11
MS-CLAP 2022	196M	0.16	2.33	27.94	14.27	89.35	25.54	5.36	53.06	50.09	52.08	13.61	10.51	28.69
CLAP-HTSAT (unfused)	153M	0.36	1.33	37.56	15.98	92.58	33.72	12.78	56.91	46.98	53.96	13.26	9.35	31.23
CLAP-HTSAT (fused)	154M	0.18	2.00	38.75	15.27	93.62	29.29	10.82	54.95	47.85	53.18	14.65	11.76	31.03
Speech encoders (no text tower; avg over the 8 non-retrieval / non-zero-shot columns [†])														
Dasheng-0.6B	630M	–	–	39.76	20.09	57.73	16.24	8.65	55.92	46.29	50.70	–	–	36.92 [†]
Dasheng-base	86M	–	–	39.04	17.22	53.91	24.10	9.13	55.94	52.38	50.70	–	–	37.80 [†]
Whisper-large-v3	1.5B	–	–	41.31	17.22	50.85	16.64	9.68	57.83	48.49	51.71	–	–	36.72 [†]
Whisper-small	244M	–	–	37.23	19.53	52.18	13.19	10.30	58.40	45.61	52.37	–	–	36.10 [†]
Omni embedders														
LCO-Embedding-Omni-7B	8.9B	83.29	74.00	36.05	16.16	70.75	43.40	0.62	53.56	<u>97.60</u>	50.82	31.67	<u>97.42</u>	54.61
LCO-Embedding-Omni-3B	4.7B	83.13	72.00	<u>31.03</u>	16.87	59.42	<u>48.97</u>	0.62	52.40	96.92	50.58	26.94	97.40	53.02
BidirLM-Omni-2.5B	2.5B	83.05	74.33	29.92	15.04	85.38	49.49	0.60	53.20	95.99	51.02	34.24	97.30	55.80
omni-embed-nemotron-3B	4.7B	<u>83.21</u>	81.33	21.35	17.12	60.45	33.05	<u>0.66</u>	53.68	98.48	<u>51.87</u>	<u>35.35</u>	97.64	52.85
e5-omni-3B	4.7B	79.69	74.67	28.93	<u>17.00</u>	79.82	37.72	<u>0.66</u>	<u>54.49</u>	73.19	51.00	33.33	93.99	52.04
e5-omni-7B	8.9B	82.80	<u>79.67</u>	30.69	14.85	<u>81.26</u>	35.89	0.87	55.40	90.30	52.13	37.43	97.03	<u>54.86</u>
gemini-embedding-2 [‡]	?	82.34	83.33	21.74	16.84	79.25	31.92	0.55	53.40	98.05	61.56	9.65	79.82	51.54
Ours: Omni-Embed-Mini														
0.9B, no mining	935M	68.84	65.00	18.91	17.05	63.03	39.87	0.63	51.66	68.37	50.57	17.99	60.97	43.57
0.9B, text mining	935M	66.84	58.67	19.77	14.61	58.76	38.01	0.63	52.03	69.26	50.26	18.61	67.97	42.95
0.9B, media mining	935M	70.19	66.00	19.35	15.91	63.74	37.26	0.58	52.02	69.66	49.79	18.68	69.83	44.42
0.9B, text + media	935M	67.67	66.00	17.99	16.67	63.77	42.16	0.64	51.73	70.56	49.94	20.97	65.02	44.43
0.9B, t+m, orig. captions	935M	57.24	63.67	18.79	17.70	56.68	40.77	0.54	51.33	73.63	50.15	13.96	56.53	41.75
2.3B, text + media	2.3B	63.90	52.67	20.20	16.93	64.87	36.79	0.88	52.43	60.49	51.47	17.36	60.68	41.56
Omni-Embed-Mini-0.9B	935M	70.42	66.33	19.30	15.07	59.10	40.71	0.57	52.22	69.13	50.41	15.56	60.53	43.28
Omni-Embed-Mini-2.3B	2.3B	81.26	68.33	18.38	15.53	63.23	43.46	0.55	51.66	87.74	49.73	14.44	92.05	48.86

Table 6: MAEB Speech per-task scores across 12 tasks. **Task abbreviations:** GigaSpeechT2A (GigaSp), Spoken-SQuADT2A (SpSQuAD), CREMA-D (CREMA), CommonLanguageAgeDetection (AgeDet), IEMOCAPGender (IEMOCAP), VoxCelebSA (VoxCel), CREMA-D-Clustering (CREMA-C), CREMA-D-PairClassification (CREMA-P), NMSQA-PairClassification (NMSQA), VoxPopuli-AccentPairClassification (VoxPop), Ravdess-Zeroshot (Ravd), SpeechCommands-Zeroshot-v0.02 (SpCmd). **Metrics** appear below each task name: HR@5 = HitRate@5, Acc = accuracy, V-meas. = V-measure, max-AP = pair-classification max average precision, ZS Acc = zero-shot accuracy. Bolding, colour and footnote conventions follow §A. [†]Speech-only encoders lack a text tower and are not scored on the two retrieval tasks or the two zero-shot tasks; their average is over the remaining eight columns.

call to Qwen3-Omni-30B-A3B-Instruct at temperature 0.7 with a 2048-token output budget. The model receives the raw media plus a modality-specific system prompt that interpolates the source dataset’s ground-truth caption as a “Reference caption” grounding prior; a short user-facing prompt is drawn at random from a per-modality bank and concatenated with the appropriate media tag.

B.1 Modality system prompts

System prompt: Audio

Describe only what you hear; no opinions, speculation, or filler. Be precise and factual. Do not hallucinate sounds not present. Listen to this audio. Reference caption: “{original_text}”. Write a dense, semantically rich description. Cover: each distinct sound and its characteristics (pitch, volume, texture); temporal order (what happens first, next, last); acoustic environment (indoor/outdoor, reverb, background); patterns or repetitions. Be concise; every word should convey information about the audio. One paragraph.

System prompt: Music (audio music subsets)

Describe only what you hear. Listen to this music. Reference caption: “{original_text}”. Write a dense description covering: instruments and timbres; melody, harmony, key, chord progressions; rhythm, tempo (BPM estimate), time signature; genre, energy, dynamics; structure; vocals if present (gender, style, lyrics gist). One paragraph.

System prompt: Speech

Listen to this speech audio. Write a single flowing paragraph covering: (1) the speaker (age group, gender, voice pitch/tone, clarity, accent); (2) speaking style (pace, loudness, emotion); (3) a transition into the transcription using a casual phrase followed by the marker [TRANSCRIPTION]; (4) audio quality, recording environment, background sounds. Use the literal text [TRANSCRIPTION]; do not write out the actual words spoken. The placeholder is post-processed into a transcription span around the source transcript.

Model	Params	VOC07	Ctry211	OKVQA	EDIS	COCO-T2I	VN-T2I	COCO-I2T	VN-I2T	NIGHTS	FashIQ	Avg
		CLS	CLS	VQA	T2I	T2I	T2I	I2T	I2T	I2I	I2I	
Vision-language embedders												
seed1.6-embedding-1215	?	89.30	46.90	74.90	96.70	81.30	83.80	77.80	83.70	70.70	50.40	75.55
seed-1.6-embedding	?	91.90	47.70	74.20	91.50	78.80	83.40	77.40	84.50	72.10	49.00	75.05
Qwen3-VL-Embedding-8B	8.1B	93.40	27.30	77.80	96.30	81.10	81.10	79.10	85.70	72.70	44.50	73.90
IFM-TTE-7B	8.3B	83.40	60.50	83.40	95.50	77.70	79.20	72.60	82.50	69.40	32.40	73.66
WeMM-Embedding-8B	8.8B	94.50	32.20	73.40	94.90	81.20	81.70	78.20	84.90	69.00	43.80	73.38
RzenEmbed-v2-7B	8.3B	91.40	29.70	73.10	97.20	81.10	82.60	78.00	85.80	69.00	41.50	72.94
RzenEmbed-v1-7B	8.3B	91.60	29.00	69.60	96.50	78.60	82.60	76.00	85.80	68.60	38.70	71.70
WeMM-Embedding-2B	2.1B	93.70	28.90	66.40	93.30	77.60	76.90	77.00	80.30	69.60	43.20	70.69
Ops-MM-embedding-v1-7B	8.3B	84.80	28.50	70.60	94.40	79.30	80.10	72.10	84.30	66.20	24.30	68.46
Qwen3-VL-Embedding-2B	2.1B	84.70	20.90	65.60	88.30	75.40	70.40	71.90	74.20	67.80	37.50	65.67
VLM2Vec-V2.0	2.2B	76.80	19.50	44.00	82.90	73.80	72.30	66.40	73.70	66.20	19.30	59.49
qwen3.5-0.8b-vision (I2I only)	100M	—	—	—	—	—	—	—	—	59.30	3.80	31.55 [†]
Omni embedders												
LCO-Embedding-Omni-7B	8.9B	56.40	33.20	59.90	66.70	64.60	62.40	61.20	61.20	62.90	20.70	54.92
LCO-Embedding-Omni-3B	4.7B	71.20	28.90	58.20	56.50	59.50	55.20	59.40	59.20	56.20	15.90	52.02
omni-embed-nemotron-3B	4.7B	53.40	23.20	16.10	82.60	60.50	61.10	52.50	59.50	64.10	8.60	48.16
BidirLM-Omni-2.5B	2.5B	68.10	8.00	20.30	72.90	72.10	48.50	63.60	30.60	62.60	13.10	45.98
e5-omni-3B	4.7B	76.80	28.30	63.20	86.10	74.90	73.00	71.60	78.60	67.10	19.40	63.90
e5-omni-7B	8.9B	84.80	32.60	71.00	91.80	76.90	77.30	74.90	81.70	67.60	25.00	68.36
gemini-embedding-2 [‡]	?	67.40	13.80	17.50	87.90	74.60	82.30	2.80	45.60	64.10	16.70	47.27
Ours: Omni-Embed-Mini												
0.9B, no mining	935M	71.30	1.50	7.00	42.00	25.60	7.60	21.10	4.70	41.50	2.50	22.48
0.9B, text mining	935M	68.40	1.50	4.50	42.50	15.10	4.40	13.40	5.00	40.10	1.90	19.68
0.9B, media mining	935M	61.50	1.60	4.00	40.60	22.90	8.30	23.10	8.10	55.40	4.70	23.02
0.9B, text + media	935M	61.40	1.00	9.60	47.70	34.40	9.60	28.90	11.50	51.00	7.20	26.23
0.9B, t+m, orig. captions	935M	69.20	1.40	1.00	33.60	7.50	3.20	8.70	3.90	45.30	2.50	17.63
2.3B, text + media	2.3B	83.80	20.90	65.00	86.70	72.70	70.90	72.30	73.20	68.00	34.50	64.80
Omni-Embed-Mini-0.9B	935M	60.60	1.60	8.80	46.00	29.40	8.70	35.40	11.10	53.90	7.40	26.29
Omni-Embed-Mini-2.3B	2.3B	83.80	20.90	65.00	86.70	72.70	70.90	72.30	73.20	68.00	34.50	64.80

Table 7: MMEB-V2 image per-task hit@1 ($\times 100$). **Task abbreviations:** VOC2007 (VOC07), Country211 (Ctry211), OK-VQA (OKVQA), EDIS, MSCOCO-T2I (COCO-T2I), VisualNews-T2I (VN-T2I), MSCOCO-I2T (COCO-I2T), VisualNews-I2T (VN-I2T), NIGHTS, FashionIQ (FashIQ). **Task families** appear below each task name: CLS = zero-shot classification accuracy; VQA = knowledge retrieval hit@1; T2I = text-to-image retrieval hit@1; I2T = image-to-text retrieval hit@1; I2I = image-to-image retrieval hit@1. Bolding, colour and footnote conventions follow §A. [†]qwen3.5-0.8b-vision supports only the two I2I tasks; its average is over those two columns and is not directly comparable to the others.

System prompt: Image

Look at this image. Reference caption: “{original_text}”. Write a dense, factual description covering: subjects and objects (what, how many, attributes); spatial layout and composition; colors, lighting, and style; any actions or interactions; any text visible in the image (OCR). Be concrete; avoid speculation about meaning or backstory. One paragraph.

System prompt: Video

Watch this video. Reference caption: “{original_text}”. Write a dense description covering: subjects and setting; spatial layout; the temporal sequence of events (what happens first, next, last); camera motion or shot transitions; on-screen actions; any text or captions visible on screen. One paragraph.

System prompt: Visual document

You are an expert document analyst. Examine this document image and provide a comprehensive description covering: document type and purpose; overall layout and structure; all text content (headings, body, labels); tables (column headers, row structure, key data); charts and graphs (type, axes, trends); figures and diagrams; visual formatting (colors, fonts, emphasis). Be thorough, precise, and well-organised.

B.2 User-prompt banks

For each modality the user-side message is sampled from a small bank of natural-language requests: e.g. “what do you hear”, “describe this audio”, “caption this” for audio; “what did they say”, “transcribe this” for speech; “what do you see”, “describe this image” for image; and “what do you see”, “describe this video”, “what is happening in this video” for video. Visual-document rows use a single short user prompt; the grounding work is carried by the system prompt.

Model	Params	HMDB	UCF	MSRVTT	MSVD	DiDeMo	VATEX	Avg
		V-CLS	V-CLS	V-T2V	V-T2V	V-T2V	V-T2V	
Vision–language embedders								
seed1.6-embedding-1215	?	93.50	98.70	60.20	74.48	66.43	54.87	74.70
Qwen3-VL-Embedding-8B	8.1B	83.40	95.10	58.20	75.67	66.04	54.87	72.21
WeMM-Embedding-8B	8.8B	61.50	84.20	55.30	73.43	69.22	55.05	66.45
WeMM-Embedding-2B	2.1B	57.70	75.10	53.10	73.43	65.24	52.52	62.85
IFM-TTE-7B	8.3B	65.40	79.60	52.70	73.13	49.70	51.45	62.00
seed-1.6-embedding	?	63.90	74.20	55.30	71.34	56.67	48.77	61.70
RzenEmbed-v2-7B	8.3B	63.40	75.70	50.60	72.09	57.27	47.57	61.10
Ops-MM-embedding-v1-7B	8.3B	59.30	78.80	49.40	67.16	46.12	43.95	57.45
Qwen3-VL-Embedding-2B	2.1B	63.60	83.50	42.50	65.67	41.33	37.74	55.72
RzenEmbed-v1-7B	8.3B	56.70	73.10	45.10	64.93	41.24	38.57	53.27
VLM2Vec-V2.0	2.2B	35.20	57.30	28.80	43.73	29.98	23.83	36.47
Omni embedders								
LCO-Embedding-Omni-3B	4.7B	39.00	<u>65.10</u>	38.40	58.06	31.47	31.13	43.86
LCO-Embedding-Omni-7B	8.9B	30.50	64.50	<u>41.50</u>	59.10	32.37	32.05	43.34
omni-embed-nemotron-3B	4.7B	22.40	36.60	35.70	55.22	32.37	27.09	34.90
BidirLM-Omni-2.5B [†]	2.5B	34.10	55.70	36.10	58.96	34.06	28.43	41.23
e5-omni-3B	4.7B	19.60	19.50	36.10	58.21	<u>36.35</u>	27.76	32.92
e5-omni-7B	8.9B	<u>44.80</u>	58.30	38.60	<u>59.25</u>	35.16	<u>32.29</u>	<u>44.73</u>
gemini-embedding-2 [‡]	?	32.20	56.30	47.50	66.12	45.82	40.89	48.14
Ours: Omni-Embed-Mini								
0.9B, no mining	935M	10.90	14.50	9.70	19.10	9.26	7.73	11.87
0.9B, text mining	935M	10.80	13.60	9.10	7.76	4.28	2.23	7.96
0.9B, media mining	935M	13.20	15.10	11.60	18.51	7.57	4.91	11.81
0.9B, text + media	935M	16.70	27.40	18.50	27.61	12.75	9.83	18.80
0.9B, t+m, orig. captions	935M	9.00	7.40	5.30	3.88	1.69	1.18	4.74
2.3B, text + media	2.3B	62.40	83.40	42.20	65.52	40.84	36.74	55.18
Omni-Embed-Mini-0.9B	935M	16.20	29.30	15.80	26.42	12.55	10.63	18.48
Omni-Embed-Mini-2.3B	2.3B	62.40	83.40	42.20	65.52	40.84	36.74	55.18

Table 8: MMEB-V2 video per-task hit@1 ($\times 100$). **Task abbreviations:** HMDB51 (HMDB), UCF101 (UCF), MSR-VTT (MSRVTT), MSVD, DiDeMo, VATEX. **Task families** appear below each task name: V-CLS = zero-shot action classification accuracy; V-T2V = text-to-video retrieval hit@1. Across all omni-style models, the benchmark’s own 8 pre-extracted frames per video are embedded as images and mean-pooled, so every model receives identical inputs under identical frame-level aggregation; the protocol may nevertheless understate a model’s native temporal capability. Bolding, colour and footnote conventions follow §A. [†]BidirLM-Omni-2.5B does not declare a video modality, but it accepts up to 8 image frames, so we evaluate it under the same 8-frame protocol.

B.3 Sample schema

Each row of the caption slice carries a unique row identifier, the media, a chat-style messages field holding the user request and the model’s dense caption, the modality label (audio, speech, image, video, or visual-doc), the caption category (always “caption” for this slice), the source dataset’s original short caption (used as the ground-ing prior), and metadata fields for the source dataset

name, sampling rate (16,000 for audio/speech) and frame rate (for video).

B.4 Qualitative dense-caption examples

Table 10 shows one representative example per modality from the caption slice. The middle column is a brief written gloss of the underlying media (a one-line description sitting between the original short caption and the dense caption); the third col-

Model	Params	Fin	HR	Ind	Phr	CS	En	Phy	Avg
<i>Visual-document retrievers</i>									
colqwen3.5-4.5B-v3	4.6B	68.69	66.09	59.35	67.31	79.87	66.16	50.08	65.36
nomic-embed-multimodal-7b	7.8B	59.43	58.27	48.60	63.44	71.47	56.96	44.70	57.55
nomic-embed-multimodal-3b	3.8B	57.84	54.88	46.63	60.35	70.29	54.96	45.17	55.73
<i>Vision-language embedders</i>									
llama-nemotron-embed-vl-1b-v2	1.7B	64.53	60.59	47.85	63.52	74.01	57.72	43.65	58.84
Qwen3-VL-Embedding-2B	2.1B	62.23	58.45	47.93	62.51	73.48	56.81	46.64	58.29
google/siglip-so400m-p14-384	878M	11.04	11.22	8.91	22.14	19.58	7.57	10.25	12.96
<i>Text retrievers (operate on OCR'd page text)</i>									
embeddinggemma-300m	308M	51.03	49.06	42.44	60.28	67.07	55.74	43.21	52.69
BAAI/bge-large-en	335M	38.71	45.71	39.39	55.81	64.44	30.48	32.58	43.87
<i>Omni embedders</i>									
LCO-Embedding-Omni-7B	8.9B	55.13	55.31	41.94	61.14	69.43	55.44	44.64	54.72
BidirLM-Omni-2.5B	2.5B	58.87	51.58	41.17	56.23	68.85	51.92	42.77	53.06
LCO-Embedding-Omni-3B	4.7B	54.40	52.06	39.37	58.99	67.38	53.86	44.31	52.91
omni-embed-nemotron-3B	4.7B	<u>63.98</u>	58.69	<u>49.80</u>	<u>64.07</u>	<u>73.57</u>	<u>59.71</u>	47.59	<u>59.63</u>
e5-omni-3B	4.7B	60.41	<u>59.91</u>	<u>49.76</u>	63.36	71.91	59.29	46.01	58.66
e5-omni-7B	8.9B	65.41	61.27	49.83	66.48	<u>73.84</u>	63.29	<u>46.86</u>	61.00
gemini-embedding-2 [‡]	?	63.12	57.88	46.91	63.52	72.04	51.06	45.13	57.09
<i>Ours: Omni-Embed-Mini</i>									
0.9B, no mining	935M	45.97	40.54	36.87	51.67	62.35	47.67	40.62	46.53
0.9B, text mining	935M	45.43	41.77	37.86	52.25	62.24	48.41	40.44	46.91
0.9B, media mining	935M	46.85	44.57	36.01	52.56	61.98	46.61	36.36	46.42
0.9B, text + media	935M	46.76	41.04	37.35	52.58	62.00	48.11	41.03	46.98
0.9B, t+m, orig. captions	935M	38.24	39.56	32.47	48.84	57.75	42.80	35.82	42.21
2.3B, text + media	2.3B	62.64	57.61	47.89	62.01	74.43	56.11	46.00	58.10
Omni-Embed-Mini-0.9B	935M	44.69	41.22	37.94	53.17	61.69	49.19	40.57	46.92
Omni-Embed-Mini-2.3B	2.3B	62.64	57.61	47.89	62.01	74.43	56.11	46.00	58.10

Table 9: ViDoRe-V3 per-domain scores. All seven tasks are text-to-page retrieval (T2Pg) over visual-document page images, scored by nDCG@10 ($\times 100$). **Domain abbreviations:** Finance (Fin), HR, Industrial (Ind), Pharmaceuticals (Phr), Computer Science (CS), Energy (En), Physics (Phy). Bolding, colour and footnote conventions follow §A. The mining ablation rows show that visual-document retrieval is largely insensitive to which side hard negatives are drawn from: the four 0.9B mining configurations agree to within 0.6 nDCG@10 (46.42 to 46.98). What does move this modality is caption quality: replacing the dense cascaded captions with the datasets’ original captions costs 4.8 nDCG@10 (46.98 $\rightarrow$ 42.21).

umn is the original short caption from the source dataset (the grounding prior); the fourth column is the dense caption produced by the cascaded captioner and used as the contrastive teacher target.

C Per-source datasets, licenses, citations

The caption slice draws from 16 unique source datasets across 5 modalities, totalling 242,080 caption rows and 54.9 days of media. Our generated dense captions are released under CC BY-SA 4.0;

the source media and their original annotations remain under the licenses of their respective datasets, summarised in Table 11.

C.1 Train/test overlap audit

Three of our training sources share a parent corpus with MMEB-V2 evaluation tasks, so we ran an identifier-level audit rather than relying on split names. For each training sample we recovered the original source identifier and compared it one

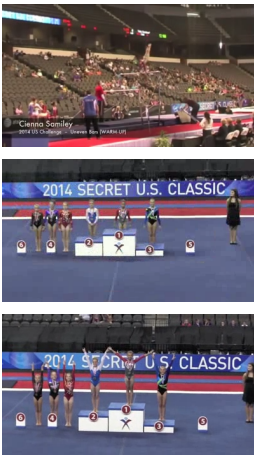
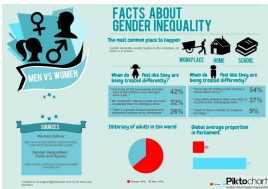
Modality	Media (described)	Original caption	Dense caption
Audio	A woman speaks in a public space while a slow heartbeat-like thump and faint ticking play in the background.	A woman talks while a heartbeat and ticking play in the background.	A woman’s voice, speaking in a mid-to-low pitch, is present at the beginning of the clip, accompanied by faint, indistinct conversation from other people. Simultaneously, a deep, rhythmic thumping with a slow, steady tempo, resembling a heartbeat, begins and continues throughout. The background includes a persistent, low-level ambient hum and the general murmur of a public indoor space with some reverberation.
Speech	A young woman with a clear, neutral accent reads a short factual question in a calm, measured voice in a quiet indoor setting.	What are organic plants understood to be?	A young adult female with a high-pitched, clear, and articulate voice delivers the line in a standard, neutral accent. Her speaking style is measured and calm, with a moderate pace and consistent volume, conveying a neutral, informative affect. She says, “What are organic plants understood to be?” The audio is clean and crisp, likely recorded in a quiet indoor space with no discernible background noise or reverberation.
Image		Two women going down a conveyor belt with a cart full of groceries.	The image, taken from a high-angle perspective, shows two women on a descending escalator. The woman in the foreground, wearing a floral-patterned short-sleeved shirt, leans forward, gripping the handle of a metal shopping cart. Behind her, another woman with brown hair, wearing a white top over a dark shirt, also holds the cart’s handle. The cart is filled with various items, including a visible bag of potato chips and other grocery products. The setting appears to be indoors, with a neutral-toned, geometric-patterned wall in the background; lighting is even and diffuse, characteristic of an indoor public space.
Video		There is a girl gymnast who is performing on the bars.	A gymnast, Cienna Samiley, performs on the uneven bars during the 2014 US Challenge warm-up. The scene is a large, indoor arena with sparse audience seating. The camera cuts to a podium where seven young female gymnasts stand, wearing leotards and medals, in front of a banner reading “2014 SECRET U.S. CLASSIC”. A scoreboard graphic is superimposed, displaying the top six scores. The gymnasts on the podium then raise their arms and cheer.
Visual doc		(not available)	A detailed analysis of an infographic titled “Facts About Gender Inequality,” which uses a combination of text, icons, and data visualizations to present information on gender disparities. The infographic is set against a light teal background, structured into several sections, with a bold sans-serif title at the top ... Icons use simple black silhouettes (gender symbols, a wheelbarrow for the workplace, a house for home, books for school); the colour scheme is teal/red/black; data is presented via a pie chart and a bar chart with a consistent red-for-women, blue-for-men encoding.

Table 10: One sample per modality from the caption slice. Visual documents in the source corpus carry no human-written original caption; the cascaded captioner generates the dense description directly from the page image, and we mark such cells “(not available)”. Visual-document dense captions are themselves substantially longer and richer than the other modalities; the example shown here is truncated for layout (indicated by “...”). In the video example, the gymnast’s name and the event are read from on-screen text in the video itself, not inferred by the captioner.

to one against the identifiers of the benchmark test samples. **The audit finds zero shared items in all three cases.**

MSR-VTT. Evaluation uses the benchmark’s test_1k split. Training and evaluation identifiers are disjoint and their union is exactly the 10,000 videos of the corpus, so the two sides are the complementary 9k/1k halves.

ActivityNet. Evaluation uses the test split, 614 unique videos after normalising YouTube identifiers; the training side contributes 10,009 videos, of which 9,793 carry dense captions. The intersection is empty on both the full training set and the captioned subset. ActivityNet is present in training but is not among the MMEB-V2 tasks we report; we audit it so that it is confirmed clean for future evaluation.

MSCOCO, via Recap-COCO-30K. Matching is on the COCO image_id parsed from the image filenames, with the evaluation images drawn from val2014. No training image identifier appears among the MSCOCO-I2T or MSCOCO-T2I test identifiers. We state this at the identifier level because that is what we verified; we make no claim about how the upstream Recap-COCO-30K release partitions the original COCO splits.

Because no overlapping items exist, no rows were removed and the reported cross-modal numbers stand unchanged.

D Mining subsystem

The miner runs as a single background thread that shares the live training model with the main loop, avoiding the roughly 16 GB extra HBM cost of a separate model copy. Within-cycle weight drift between miner and main thread is empirically negligible because the LoRA delta per optimiser step is small and the embedding geometry is dominated by the frozen backbone.

Cadence. Every modality receives mined hard negatives. The miner alternates between two cycle types, gated on the training step counter. A *text* cycle re-embeds the captions of all five media modalities, builds a CPU FAISS index, and writes hard-negative ids into the negative cache; it fires roughly ten times per epoch and finishes in about thirty seconds. A *media* cycle re-embeds the samples of each media modality (speech, audio, image, video and visual document) through the full model into per-modality indices; it fires roughly five times per epoch and takes 12 to 20 minutes. Triggers are

coalesced, so multiple firings during a long cycle collapse to a single follow-up.

Concurrency and cache semantics. A single forward-lock serialises miner and main-thread forward passes at batch granularity: the main thread holds the lock for the entire forward, backward and optimiser step, and the miner acquires it per encode-batch, yielding fine-grained alternation rather than overlap. The negative cache is keyed by sample id and overwrites on write, so media-mined entries supersede earlier text-mined ones for the same sample. If the cache is empty for a given sample at batch time the collator falls back to in-batch negatives plus distillation, recovering the no-mining ablation behaviour with no special-case code.

E Caption grounding and noise safeguards

Because every teacher target is machine-generated, caption noise is a direct risk to the learned geometry. Four properties of the recipe limit how far that noise can propagate. First, the captioner is *grounded*: the source dataset’s own caption is interpolated into the prompt, so the model elaborates on a human-written anchor rather than describing the media unconstrained. Second, the target is a *pooled embedding*, not the caption tokens; the student fits a single vector, so local errors in wording are attenuated by pooling rather than supervised token by token. Compression is not correction, however: systematic hallucinations, omissions and semantic biases can still propagate into the student through the pooled target. Third, hard-negative mining applies a 0.92 cosine cutoff, which discards candidates too close to the anchor to be genuine negatives and so removes the mining pairs most likely to be caption artefacts. Fourth, the text path is frozen, so no amount of captioner bias can reach text retrieval: whatever the captions contain, the text branch is the stock backbone.

Visual documents are the informative case: they are the one modality captioned with no reference caption to ground against, and they still reach strong retrieval, which shows that fully generated captions can provide effective supervision in this setting. It does not measure robustness to hallucinated or systematically biased captions. These are containment properties, not verification; we do not filter captions for factuality, and the Limitations section states what that leaves unmeasured.

Source	Modality	Samples	License
TED-LIUM 3 (Hernandez et al., 2018)	speech	26,826	CC BY-NC-ND 3.0
HeySQuAD (Wu et al., 2023a)	speech	7,199	CC BY 4.0 (HF declared); SQuAD text CC BY-SA 4.0
Spoken Alpaca (Shih et al., 2023)	speech	5,135	CC BY-NC 4.0
Gemini Speech (SB, 2025)	speech	4,726	Apache 2.0
LibriSpeech (Panayotov et al., 2015)	speech	2,854	CC BY 4.0
WavCaps (FreeSound) (Mei et al., 2024)	audio	26,217	Academic / research only
WavCaps (AudioSet-SL) (Mei et al., 2024)	audio	10,831	Academic / research only
AudioCaps (Kim et al., 2019)	audio	4,518	MIT (academic only)
WavCaps (BBC) (Mei et al., 2024)	audio	3,115	Academic / research only
Clotho (Drossos et al., 2020)	audio	394	Tampere Univ. (non-commercial); audio: per-clip Freesound CC
WavCaps (SoundBible) (Mei et al., 2024)	audio	118	Academic / research only
ShareGPT-4o (image) (Chen et al., 2024)	image	42,271	MIT (academic only)
Recap-COCO-30K (Li et al., 2024)	image	22,474	CC BY 4.0
Flickr30k (Young et al., 2014)	image	21,398	Research-only (Flickr ToU)
ActivityNet (Krishna et al., 2017)	video	27,526	No explicit license (devkit MIT); YouTube ToS
MSR-VTT (Xu et al., 2016)	video	6,612	Research-only (Microsoft)
ShareGPT-4o (video) (Chen et al., 2024)	video	1,555	MIT (academic only)
VisRAG (Yu et al., 2025)	visual doc	14,995	Mixed (per sub-source)
ColPali (Faysse et al., 2025)	visual doc	13,316	Mixed (per sub-source)

Table 11: Per-source breakdown of the caption slice of our omni-modal corpus, with citations and source licenses. Our generated captions are released under CC BY-SA 4.0; source media keep the licenses listed here.

F Additional ablations

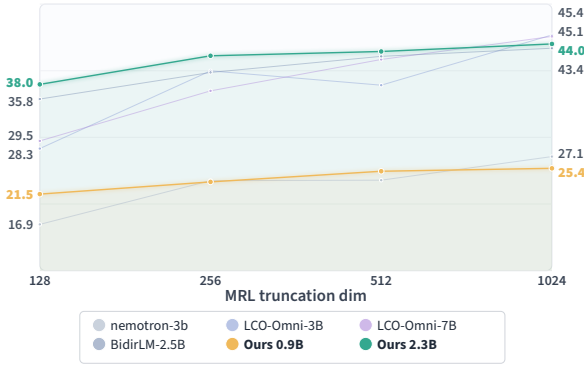


Figure 8: **MRL truncation behaviour.** AVG over per-modality probes (two retrieval tasks each from MTEB-v2, MAEB, MMEB-V2, ViDoRe-V3) against truncation dimension $d \in \{128, 256, 512, 1024\}$. Both *Ours* curves are near-monotone and stay close to full-dimension performance even at the most aggressive 128-dim slice, whereas the omni-specific baselines fall away far more steeply and their per-modality curves are not monotone. Vision-language embedders that ship their own Matryoshka training degrade comparably gently.

Matryoshka truncation. Fig. 8 plots the truncation curves summarised in §6.4. The additional analyses reported below are a SciFact text-vs-speech NDCG@10 convergence probe, emergent media-to-media alignment via the shared text axis, and the single-stage data-loader with its phased optimiser schedule.

Seed sensitivity, per seed. Table 12 gives the per-seed scores behind the summary in §6.3. All three

runs use the 10-epoch no-mining configuration of the 0.9B variant and differ only in the random seed.

Modality	s42	s71	s1234	Mean	SD
Text	49.57	49.57	49.57	49.57	0.00
Speech	43.93	41.95	43.41	43.10	1.03
Audio	33.28	34.22	34.37	33.96	0.59
Image	27.96	27.35	27.13	27.48	0.43
Video	20.86	19.61	18.99	19.82	0.95
Visual Doc	46.76	46.21	43.47	45.48	1.76

Table 12: **Seed sensitivity** of the 0.9B variant across seeds 42, 71 and 1234, on the 10-epoch no-mining configuration. Metrics match the main tables. SD is the sample standard deviation over the three seeds; the text row is identical by construction.

Cross-modal alignment probe. During training we monitor a SciFact text-vs-speech NDCG@10 convergence probe every 10% of an epoch. The synthesised speech version of each claim is encoded through the full multimodal path (Whisper + Dasheng $\rightarrow$ projectors $\rightarrow$ backbone) while the corpus remains text. The two NDCG@10 curves converging is the headline indicator that the audio path has reached the backbone’s text geometry.

Media-to-media alignment. Routing every modality through the *same* shared backbone and aligning each one to the caption text places all media in one geometry after training. We never train a media-to-media objective, but because every modality shares the text axis as a common reference frame, pairings such as image $\leftrightarrow$ audio, audio $\leftrightarrow$ video and image $\leftrightarrow$ document can in

principle be compared directly. This mirrors the emergent alignment reported by ImageBind (Girdhar et al., 2023), here induced by a text anchor rather than by contrasting modalities pairwise, with the frozen text-embedding backbone supplying the shared metric.

Single-stage training. LLaVA (Liu et al., 2023) and most VL models train in two distinct stages with two distinct datasets. Omni-Embed-Mini runs as a *single-stage* pipeline: the dataset is loaded once and the only “phasing” is internal to the optimisation schedule (projectors only, then projectors plus encoder LoRA; §3.1). The two phases share the same data-loader, the same dense cascaded captions, and the same loss, with no hand-off. This is possible because dense cascaded captions carry both alignment signal and task signal in one row.

G Loss equations

We give the full form of the Matryoshka SigLIP contrastive loss summarised in Sec. 3.4. Let $\hat{a}_i^{(d)}$, $\hat{p}_j^{(d)}$, $\hat{h}_{ik}^{(d)}$ denote the L2-normalised, d -truncated student, in-batch positive, and hard-negative embeddings, the last being the k -th negative mined for anchor i . The in-batch positive is the teacher embedding, so $\hat{a}_i \equiv \hat{s}_i$ and $\hat{p}_i \equiv \hat{t}_i$. With learned temperature $\tau = \exp(\log \tau)$ (clamped to $[0.01, 0.3]$) and learned bias b initialised to -10 , both shared across d ,

$$z_{ij}^{(d)} = \frac{1}{\tau} \hat{a}_i^{(d)} \cdot \hat{p}_j^{(d)} + b, \quad y_{ij} = \begin{cases} +1 & i = j \\ -1 & i \neq j \end{cases}, \quad (1)$$

$$\zeta_{ik}^{(d)} = \frac{1}{\tau} \hat{a}_i^{(d)} \cdot \hat{h}_{ik}^{(d)} + b, \quad (2)$$

and the per-dim SigLIP loss is the mean over all concatenated pairs:

$$\mathcal{L}_{\text{siglip}}^{(d)} = -\frac{1}{B(B+K)} \left[\sum_{i,j} \log \sigma(y_{ij} z_{ij}^{(d)}) + \sum_{i,k} \log \sigma(-\zeta_{ik}^{(d)}) \right] \quad (3)$$

The Matryoshka aggregation uses $1/\sqrt{d}$ weighting, $w_d = (1/\sqrt{d}) / \sum_{d'} (1/\sqrt{d'})$, upweighting aggressive truncations to reflect their higher per-coordinate retrieval cost. The cosine distillation term is $\mathcal{L}_{\text{dist}} = \sum_d w_d \mathbb{E}_i [1 - \hat{s}_i^{(d)} \cdot \hat{t}_i^{(d)}]$, and the total loss is $\mathcal{L} = \mathcal{L}_{\text{contr}} + \lambda \mathcal{L}_{\text{dist}}$ with $\lambda = 1$.

H Hyperparameters and training schedule

Table 14 lists the hyperparameters of the smallest variant, Omni-Embed-Mini-0.9B. Training runs on 8 GPUs with DDP, BF16 mixed precision, AdamW, and a single cosine schedule with warmup. Batch

size is held near-uniform across modalities (image, visual document, audio and speech at 32; video at 8 due to its patch-token footprint).

Every number we report for a given variant comes from a single checkpoint of that variant, evaluated across all six modalities.

I Additional Limitations

- **Vision modes diverge between the 0.9B and 2.3B variants.** The 0.9B variant uses a ViT extracted from the vision-native Qwen3.5-0.8B model because the 0.6B text backbone has no vision module; the 2.3B variant reuses its backbone’s native vision module. The training configuration accommodates both, but cross-variant ablations of the two vision strategies are left to future work.
- **Visual-document captions are roughly six times longer than other modalities.** Visual-document dense captions average about 4,157 characters versus about 582 for audio. The frozen backbone’s 2,048-token context truncates the long tail of visual-document captions; truncation is consistent between teacher and student paths so cosine alignment is unaffected, but the upper end of visual-doc detail is lost from the contrastive signal.
- **Video uses per-frame mean-pooled embeddings on MMEB-V2.** For all omni-style models, the benchmark’s own 8 frames per video are embedded as images and mean-pooled. This keeps results directly comparable but caps video performance below what a native temporal encoder could reach.

Component	Params
Frozen text backbone (Qwen3-Embedding-0.6B)	595.78 M
Vision encoder (ViT from Qwen3.5-0.8B)	100.59 M
Whisper-small encoder	88.15 M
Dasheng-base encoder	85.46 M
Projectors and pooler	65.37 M
Total at inference	935.35 M

Table 13: **Per-component parameter breakdown** of Omni-Embed-Mini-0.9B, expanding its total in Table 1. Training updates the projectors and pooler plus 2.95 M of media-encoder LoRA, 68.32 M in total. The 2.3B variant replaces the backbone and its separate vision encoder with Qwen3-VL-Embedding-2B and its native vision path (2126.99 M), keeps the same audio encoders, and uses larger projectors (139.61 M), giving 2440.21 M at inference.

Component	Setting
<i>Architecture</i>	
Backbone, frozen	Qwen3-Embedding-0.6B
Hidden size	1024
MRL dimensions	{128, 256, 512, 1024}
Vision encoder	ViT extracted from Qwen3.5-0.8B
Image resolution	224×224
Video tokens / max frames	196 / 96
Audio encoders, both variants	Whisper-small encoder + Dasheng-base
Tokens per audio encoder	128
Max audio duration	60 s
<i>Optimisation</i>	
LoRA rank / α / dropout	16 / 32 / 0.0
LR, audio adapters	2×10^{-4}
LR, vision adapters	1×10^{-4}
LR, encoder LoRA	1×10^{-4}
Optimizer	AdamW, weight decay 0.01
Warmup	2% of total steps
Max gradient norm	1.0
Max sequence length	2048
Batch size per GPU, image / visual-doc / audio / speech / video	32 / 32 / 32 / 32 / 8
Epochs	2, for the reported ablation runs
Mixed precision	BF16
GPUs	8, DDP
<i>Loss and schedule</i>	
Distillation loss weight	1.0, cosine
Initial temperature	0.07, learnable
Phase-2 switch	20% of total steps
<i>Hard-negative mining</i>	
Text-mining refresh	$10 \times$ per epoch
Media-mining refresh	$5 \times$ per epoch
Text index size	50,000 captions
Media index size	10,000 per modality
Hard negatives per sample	5
False-negative threshold	0.92, cosine

Table 14: **Hyperparameters** for the Omni-Embed-Mini-0.9B variant. The 2.3B variant differs in backbone, using Qwen3-VL-Embedding-2B with native vision, and keeps the same Whisper-small and Dasheng-base audio encoders. Proportional batch sizes are adjusted, while optimisation settings remain unchanged.